

Chapter-1

Introduction

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1.1 Background

Most living being on Earth unites to form groups with others of their kind to ensure their survival. These groups are formed based on shared qualities and a common purpose. We human, as being social animals, associate ourselves to one or more groups or social categories in order to meet our individual and collective needs. Constantly interacting with members of the many different communities where we share our world is an integral part of our existence. People's sense of self is formed through their identification with a social group or in-group, as opposed to an opposing social group or out-group. When a specific social categorization attains significance within a particular social context, individuals tend to adopt a group-oriented mindset and exhibit affinity towards other members belonging to the same category. This can happen in solitary situations, such as when someone is debating whether or not to show support for her group by signing a petition or opposition to other groups, in dyadic interactions, or social events with a large number of people from different groups. Our ideas, feelings, and actions in the context of our surroundings are heavily impacted and frequently compelled by the group(s) or social categories to which we identify.

Social categorization is a required and universal human social behaviour that occurs on a variety of traits, including race, gender, sexual orientation, and political philosophy (Lamont & Molnár, 2002). On the one hand, social categorization is critical to the development of a satisfying feeling of community as well as a good sense of one's own identity (Tajfel & Turner, 1986). On the other side, it can fuel societal tensions by establishing a mentality of "us" vs "them" and affecting the access that particular groups have to economic and social resources. This could be a result of a combination of factors (Ashmore et al., 2004; Bremmer, 2018; Roccas et al., 2008; Thoits & Virshup, 1997). For example, the disparities between White and Black Americans have resulted in instances of discrimination and segregation persisting beyond the abolition of slavery, (Fox & Guglielmo, 2012). Additionally, fundamental activities of categorizing and identifying affect how individuals see and judge not only themselves but also one another and the world around them.

People in India and around the world have seen a cultural transition to multiculturalism in the twenty-first century as a result of globalization, westernization, educational and technical revolutions, and the rise of the internet. As a direct consequence of this, nations are being compelled to accommodate a continuous and ever-increasing influx of migrants hailing from a variety of racial, cultural, and religious origins. We are presented with persons who appear to be different from ourselves. It should come as no surprise that societal shifts like migration are not the only means through which new people are integrated into our social surroundings. We are already accustomed to distinguishing between individuals according to factors such as their gender, sexual orientation, or nationality due to social categorization. These “others” infiltrate our lives in a variety of ways; therefore in order to progress we must interact with others that are unique from ourselves. We build conceptions of them based on our experiences and decide whether or not to interact with or avoid them.

When individuals categorize themselves and others into in-groups and out-groups, as well as when they identify themselves with their own group, they engage in intergroup behaviour and display intergroup relationships, which can either, be positive or negative. Intergroup relationships can be positive when in-group members help out members of the out-group and consequently develop intergroup harmony. But when individuals tend to display behaviours like prejudice and discrimination, then intergroup relationships become negative in nature. Social categorization and intergroup relationship processes have a considerable impact on the cognitive, behavioural, social, and motivational processes that are involved in the formation of an individual's psyche. Consequently, these are the processes that need to be investigated in the current globalized and multicultural society because of the significance of their impact at macro and micro level within the society.

1.2 Categorization

Categorization plays a fundamental role in human cognition across various domains of experience. Many cognitive theorists have pointed out the significance of a person's ability to generalize and engage in inductive reasoning when confronted with unfamiliar situations, as it is essential for survival (Ahn et al., 2000; Bloom,

1998; Markman, 1989; Mervis & Rosch, 1981; Murphy, 2002; Rosch & Levitin, 2002).

According to Bruner et al. (1956), "to classify is to render discriminable diverse things equivalent, to organize the objects, events, and people around us into classes, and to respond to them in terms of class membership rather than uniqueness" (p. 56). People can respond more habitually in a diverse and possibly vast world by grouping things and events together in their minds because it provides commonalities on different dimensions; hence, categorisation serves as an anchor in this complex world (Lingle et al., 1984). Categorization is the process of determining how items and occurrences are comparable, equivalent, and distinct from one another, and its concept is widely investigated by researchers from a variety of fields, including linguistics, philosophy, and cognitive and social psychology. Categorization is often used interchangeably with classification and grouping; however, classification refers more accurately to the formal, behavioural act of dividing things, while categorization refers more accurately to the more intangible, psychological process of distinguishing people, events, and experiences.

1.2.1 Conceptual Definitions of Categorization

Categorization can be thought of as a combination of three factors, including background knowledge (i.e., schemas, expectancies, stereotypes), Perceived equivalence (how entities are perceived to be similar to some other entities but distinct from others) and category use (i.e., explicit labelling). These three components are constraints on each other. Thus, explicitly employing a category label such as "black" or "male" may change the degree to which we regard people as equal. This, in turn, can impact our understanding of relationships so that after adopting or forming an explicit category label, we may not only learn to regard those persons as equivalent but also grow to know that blacks and males are similar in specific ways. This knowledge may limit our future views of equivalence and labelling. In a sea of extensive current information and literature, the concept of categorization has been described traditionally in four ways using theories and empirical data.

1.2.1.1 Categorization as Representing. The process of scanning any external inputs of stimuli into internal representations is called categorization under representing definition (Macmillan et al., 1977; Medin & Smith, 1984; Mervis &

Rosch, 1981; Smith & Medin, 1981). This definition is based on the strong premise that any form of inception of inputs of stimuli as "something" commences categorization. This includes a person's belonging to a social group (e.g., this person wearing a white coat is a doctor), a person's identity (e.g., this person is Dr. Ankit), or a person's traits as viewed by perceivers. Thus, categorization is regarded as a fundamental cognitive function under this definition.

1.2.1.2 Categorization as Dichotomizing. Under this concept, it was suggested by Tajfel (1969) that people tend to assimilate their impression of others in two ways: first, in terms of "attributes which may fluctuate on continuous dimensions," and second, in terms of "classification [i.e. categories] which are discontinuous." "Nationalities or ethnic groups are, on the whole, discontinuous," he said, "whereas personal attributes or characteristics can be scientifically treated as dimensions much like height and weight would" (Tajfel, 1969, p. 178). In modern studies, discrete representations have evolved as binary representations (e.g., whether a person is religious or not), whereas non-discrete representations in the form of graded representations are used (Blair et al., 2002, 2004, 2005). Categorization has been defined as binary mapping under this approach; hence graded mapping does not represent categorization (Klapper et al., 2017).

1.2.1.3 Categorization as Organization. Categorization has been defined by a group of researchers as a method of representing others as wholes rather than as individual features; for example, a social perceiver might consider or map a doctor as a whole, which organizes the individual features of a doctor like a white coat, stethoscope, and clinical setting. This categorization criterion was used by Fiske et al. (1987) and Fiske and Neuberg (1990) proposed in a continuum model, which was widely quoted by others and developed as a significant study. They discussed two forms of covert mappings in the continuum model: "categories" and "attributes" (Fiske et al., 1987; Fiske & Neuberg, 1990) "the Features [/attributes] that an observer employs to arrange and interpret the remaining features create the category [...]" (Fiske & Neuberg, 1990, p. 9). As a result, a "category" (for example, doctor) is dominant, while all other characteristics are called "attributes". Further, they stated that an observer processes information about another person in two ways (Fiske & Neuberg, 1990). The first is traditionally known as "categorical" processing, in which the attendee engages in categorical processing of another person and processes

in terms of best available organizing representations (e.g., doctor) along with its associated preconceptions; the second is non-categorical processing, typically known as "attributes-based" or "individuating" processing, in which all apparent characteristics of another person are processed (e.g., white coat, stethoscope, and clinical setting).

1.2.1.4 Categorization as Grouping. According to this definition categorization is classifying others based on the social categories they represent rather than considering them as distinct persons (Mason & Macrae, 2004). In other ways unlike non-categorical processing, which includes mapping each individual onto a separate representation (such as "Dr. Ankit", "Dr. Shiba", and "Dr. John"), "categorization" involves mapping many people onto the same internal representation (such as "doctor", "doctor", and "doctor"). This distinction has connections to the enormous research on social identity and self-categorization (Brown, 2000; Hornsey, 2008; Tajfel & Turner, 1986), and frequently appears in the contemporary literature on person perception (Hugenberg et al., 2010; Macrae & Bodenhausen, 2000; Macrae & Bodenhausen, 2001; Mason & Macrae, 2004).

1.2.2 Functions of Categorization

There are two distinct and important views on functions of categorization:

1.2.2.1 Bruner's Sense-Making Process. Bruner (1957) established veridical perception principles, many of which exemplified categorization as a sense-making process. He described it in terms of the following properties that determine sense-making role of categorization in perception:

- a. The central property of perception is that it is a decision-making process because it entails identifying what a stimulus is.
- b. People use stimulus information to differentiate between stimuli throughout the perception judgment process and organize them into meaningful categories that best fit their perceptual set.
- c. From the detection of stimuli to the inference of identification, there is a chain of inference. The processes that make up this cue utilization chain are as follows: isolating an item from background stimulation; cue searching, which involves fitting cues to category criteria; tentative categorization, which involves more cue searching; and final categorization, which involves reduced

cue searching. A watch collector looking for a rare royal watch in the market, for example, would first differentiate a royal watch shop from the other shops in the market. The next stage could involve identifying the thing as a royal watch from other watches and looking for broad indicators that would place it in the royal watch group category. Prior to making the definitive categorization of whether this is or is not a royal watch, the initial categorization would be supplemented by conducting additional cue searches (including, for example, price and logo, standard markings).

- d. Category can be understood as a set of criteria that determine which stimuli can be grouped together as equivalent.
- e. Categorizations vary in their accessibility or perceptual readiness. Perception is compatible with former experiences and current intentions when there is a fluctuation in perceptual readiness. Most of the time, all watches are equivalent to most of us, but this is clearly not the case for the royal watch collector.
- f. In order to form inferences about the objects other features, veridical perception requires the encoding of stimulus inputs in such a way that one can move from a cue to a categorical perception. In order to accomplish this, perception needs to learn categories and category systems specific to the events and objects of interest. To generalize the nature of things and events beyond their immediate context and make accurate predictions about future occurrences, perception serves as a representational function.
- g. The veridicality of perception is determined by the compatibility of relevant categorizations with the objects and events that the individual is expected to observe. This also means that perception has different degrees of veridicality. It does not accurately portray the world.

1.2.2.1 Allport's (1954) Analysis. Allport came up with a strong functional explanation for the categorization process, arguing that categorization is done because we inherently desire to reduce the information overload generated by processing information without engaging in the categorization process. Thus we want to simplify the things through reduction of information.

- a. Because we acquire new experiences every day, we need categorization to form larger groupings and clusters that can guide us and help us adapt. Being

open-minded won't help here because we process new information in relation to our pre-existing schemas.

- b. Each piece of information is integrated into the cluster as thoroughly as possible during the categorization process. In daily situations, we categorize things and situations into the broadest possible categories.
- c. Despite making mistakes, one of the categories' primary functions is to help us perceive and act in the world by allowing us to rapidly recognize items that fit within those categories.
- d. All the items in the category share the same mental and emotional quality. Ideas, according to Allport (e.g., "Car"), are intellectual categories; concepts with emotional associations or connotative meaning include membership to a certain group.
- e. Categories with a lower likelihood of being true (irrational categories), as Allport suggests, form just as easily as categories with a greater likelihood of being true (rational categories). As a result, a category can be more or less rational.

1.2.3 Categorization to Social Categorization

A systematic manner of thinking is essential to efficiently handle the variety of stimuli in everyday life since comprehending every single detail that surrounds us is merely too expensive for our cognitive system (Lippmann, 1922). Categorization is how we attempt to understand the world around us by classifying or organizing things. We learn about relationships between objects when we identify which categories they belong to. The necessity of categorization becomes particularly apparent when engaging with the social aspects of our environment. In the absence of categorization, each individual and social situation would need to be approached as a unique entity, resulting in the social observer experiencing confusion and being flooded with an excessive amount of information that must be comprehended prior to making any decisions or undertaking any actions. Applying the same technique to people, including ourselves, is what social categorization means. It is not only a mental process for knowing and describing the world; it is also an aspect of how we arrange the world. That is, the groups to which we belong, such as genders, races, sects, and countries, are based on social categories, and consequently processes like stereotyping and person perception are influenced by social categorization.

Followers of Tajfel and others can be said to be part of the "social categorization tradition," "social identity approach," or "European intergroup approach," all of which describe a common theoretical framework for their own bodies of work. According to Tajfel (1981), one of the ways in which we develop our own "construction of any specific social reality" is through the use of social categorization. Tajfel (1978) suggested that social categorization aids in "structuring the causal understanding of the social environment" as a guide to behaviour. It also offers an orientation system for self-reference, establishing and defining the person's place within society. Social categorization gives birth to a wide variety of social processes. Examples include stereotypes (Allport, 1954) and identity (Tajfel, 1981).

1.3 Social Categorization

Social categorization refers to the process of assigning labels to oneself and others, thereby categorizing individuals into distinct social groups. This categorization is based on common characteristics, which may include factors such as ethnicity, physical attributes, or even psychological traits (Abrams, 2012; Kunda & Spencer, 2003; Stangor et al., 1992). Thus, social categorization refers to the finding that people cognitively categorize others based on their group membership (Allport, 1954; Bruner, 1957).

Considering the following scenario—a person notices someone moving across the street. Rather than focusing on every feature of this moving person, he might immediately identify him or her based on gender. As a response, the person will immediately activate his basic knowledge linked with the gender group to which that moving person belongs (Devine, 1989), that influence person's future perception and behaviour. Categorical thinking serves to simplify the complexities of the world, resulting in a significant reduction in the volume of information that necessitates evaluation (Allport, 1954; Brewer, 1988; Lippmann, 1922).

There are two basic perspectives on why people make sense through social categorization: The first relates to Taylor and Fiske's (1978) study in that people categorize to conserve limited cognitive resources by simplifying. That is, rather than remembering and storing all of the details needed to handle different aspects of the world as distinct exemplars, it is easier, faster, and simpler to group things and engage

in generalities. The consequence of such generalization is confusion and inaccuracy, yet it is better to cognitive overload. The second or alternative approach, which is related to the work of Oakes et al. (1994), is that in social perception, individuals are confronted with too little information instead of too much. Bruner argues that the strength of categorization comes in its capacity to go further than the presented information. Under this viewpoint, generalizations are not information-losing simplifications but rather additions to the stock of information that result from identifying associations that, once established, allow deductions to be made. Over the years, these perspectives have been in a heated debate concerning issues like individual bias; out-group homogeneity, and the development of stereotypical attitudes (McGarty, 1999). The first cognitive miser approach is still considered mainstream and is supported by most of the social psychology literature.

Most models that examine the social categorization process distinguish between three phases: (a) the perception of the person, (b) the activation of the category, and (c) the application of the category (Allport, 1954; Macrae & Bodenhausen, 2000). To be categorized, a person must first be perceived. Following that, important cues associated with a certain social category are identified, often using minimal cues, and categorical processing is initiated (Cloutier & Macrae, 2007). Moreover, individuals have a tendency to employ category-based processing with greater frequency when their cognitive capacity is restricted and the execution of detailed processing becomes arduous (Macrae & Bodenhausen, 2000; van Knippenberg et al., 1999). At the second level, we're involved in activating a social category and, simultaneously, our stored information about that category (Devine, 1989). Assume that, we see a person in the metro who appears to be a thief. Based on our knowledge of thief conduct, words like untrustworthy, tactless, and unpleasant may become more accessible in our minds. The activated category and related knowledge are applied to the individual we viewed as a thief in the third step. As a result, this process has an impact on how we act and interpret thief behaviour. This explains why we interpret the thief's behaviour as suspicious and motivated by a need to avoid being in closer contact to this individual.

The activation of a category and related concepts aids in the evaluation of predicted behaviour while also preserving mental energy. These resources could be used to process unusual behaviour (Sherman et al., 1998). Consequently, there are

numerous advantages to this social categorization process; it minimizes complexity and assists in the conservation of cognitive capacity that may be valuable during social contact.

1.3.1 Development of Social Categorization

Social interaction is a fundamental aspect of human existence. Human infants demonstrate a heightened level of awareness regarding their social environment. They rapidly acquire or possess innate cognitive abilities that allow them to comprehend the goals, intentions, and other mental states of individuals in their vicinity (Hamlin et al., 2008; Kuhlmeier et al., 2003; Rhodes & Baron, 2019; Spelke et al., 1995). The comprehension of the social world is significantly enhanced by the examination of collective expressions exhibited by groups of individuals. The concept that human beings transcend individuality and can be classified into different categories can be traced back to Plato's *The Republic*, published around 380 BC. In this work, Plato delineates three distinct groups to which individuals can belong: warriors, workers, and philosophers (Plato, 1943). Throughout history, the practice of classifying humans into social groups has been pervasive, encompassing various aspects such as the primary differentiation between kin and non-kin, the division of labor and social roles, and the broader establishment of social hierarchies. In order to navigate the intricacies of the social fabric, it is imperative for children to develop the cognitive ability to engage in rational thinking pertaining to the fundamental social divisions that hold significance within their respective societies.

In order to achieve this goal, children are required to overcome a series of essential obstacles in the realm of learning. In the initial stages, individuals rely on perceptual and conceptual learning mechanisms in order to acquire pertinent strategies for categorization. Furthermore, it is imperative for individuals to interpret the meaning behind these categories. Both tasks are considered to be challenging. Individuals are classified based on various factors, such as ethnicity, gender, age, physical attributes, indicators of performance, cognitive and physical abilities, geographical location, political beliefs, as well as more conceptual dimensions like affiliations, alliances, and both essential and arbitrary shared characteristics (Astuti et al., 2004). Nevertheless, these classifications exhibit variations in their level of unity, consistency across different situations, implications for the characteristics or actions

of individual members within the group, and their influence on the attitudes and behaviours of children towards individuals who belong or do not belong to the same groups as them (referred to as in-group and out-group members). Nevertheless, the particular classifications that an individual or community may be perceived to represent vary across cultural settings and over the course of human civilization.

Basic perceptual, conceptual, and social processes lay the groundwork for the complex adult systems of social categorization. Therefore, it is important to investigate the perceptual, conceptual, and social processes that contribute to the development and evolution of children's understandings of social categories over time.

1.3.1.1 Perceptual Processes. The process of categorization in various domains of human experience commences with the act of perceiving and directing attention towards pertinent cues that facilitate the identification of categories (Quinn, 2011). Race and gender have been the subjects of most research into the age at which new-borns first become aware of and begin to pay attention to variations between persons that correspond to conventional markers of social categories. Although infants do not appear to perceive racial differences in their visual environments when they are new-borns, by the time they are three months old, they begin to pay attention to the aspects of human variation that relate to traditional race categories (Kelly et al., 2005; Liu et al., 2015). For example, Bar-Haim et al. (2006) observed that three-month-old White babies in Israel spend more time staring at White faces, Black babies in Ethiopia spend more time looking at Black faces, whereas Black new-borns in Israel (who frequently view both White and Black faces) did not show any racial discrimination in their looking behaviour (Kelly et al., 2007). The recognition of gender by infants demonstrates the importance of familiarity in structuring these early processes; for example, during the initial months of infancy, newborns have a tendency to engage in longer periods of gazing at female faces, particularly when these faces belong to individuals of their own race (Quinn et al., 2008). However, this preference is subject to variation based on the gender of the primary caregiver for the infant (Quinn et al., 2002). The phenomenon of infants displaying a preference for prolonged gaze at faces of one race compared to another does not inherently imply the presence of racial categorization. It is possible that infants exhibit longer gaze durations due to perceiving perceptual distinctions between individuals, without

necessarily attributing these perceptual traits as indicative of distinct categories (Ellis et al., 2017; Liu et al., 2011; Quinn et al., 2016).

By the period they're six months old, most babies have already begun to categorize human faces into more specific types; for example, they can tell the difference between an attractive and an unattractive face at this period (Ramsey et al., 2004). During this developmental period, infants also acquire the ability to classify visual stimuli according to their emotional expressions (Kotsoni et al., 2001). Infants aged between seven and ten months demonstrate the ability to categorize individuals according to conventional gender identities. The study conducted by Leinbach and Fagot (1993) demonstrated that infants at the age of nine months, after being habituated to female faces, exhibited renewed attention when presented with unfamiliar male faces. However, their attention did not significantly increase when exposed to unfamiliar female faces. This finding suggests that infants in this age group possess the ability to categorize faces into distinct gender categories. Based on habituation frameworks, it has been observed that infants begin to engage in the process of categorization based on ethnicity at the age of six months. Notably, substantial developmental transformations take place within the initial year of childhood. At approximately six months of age, infants demonstrate the ability to categorize faces based on their racial background, particularly when they belong to the ethnic majority in their environment. However, they have not yet developed the capacity to differentiate faces originating from various ethnic backgrounds (Anzures et al., 2010).

According to Quinn et al. (2016), infants begin to develop categorical distinctions for unfamiliar racial faces at approximately nine months of age. By around twelve months of age, they may begin to differentiate between faces from their own racial group and those from other racial groups, often grouping the latter into a single category. Perceptual processing seems to be influenced by mechanisms rooted in familiarity, making it significantly influenced by the environmental factors that infants are exposed to, such as the racial and ethnic diversity within their family and community, as well as the genders of their caregivers. Additionally, the structural elements of their surroundings, including social cues related to competition and cooperation, also play a role in shaping perceptual processing (Ferera et al., 2018). One potential perceptual consequence of the emergence of these new categories

appears to be a diminished ability to distinguish individuals who are grouped together as members of a less familiar group. An instance of the phenomenon known as the other race effect, wherein individuals exhibit superior recognition abilities for individuals of their own race compared to those of other races, has been observed during infancy (Kelly et al., 2007). Furthermore, this effect has been found to be contingent upon contextual factors (Anzures et al., 2013).

Lee et al. (2017) suggest that heightened exposure to own-race faces, coupled with predominantly positive encounters such as cheerful facial expressions, contributes to the development of implicit racial bias. Nevertheless, the extent to which these effects extend remains ambiguous, such as whether perceptual experience alone has the potential to influence other forms of prejudice, such as explicit or behavioural biases. Moreover, it is important to note that there exists a substantial temporal delay between the emergence of observable racial prejudice in attitudes, beliefs, or actions and the onset of these cognitive processes during infancy (Kinzler & Spelke, 2011; Mandalaywala et al., 2019). This underscores the importance of supplementary developmental mechanisms, beyond initial visual biases, in shaping the development of more complex (and detrimental) manifestations of social bias.

1.3.1.2 Conceptual Processes. Individuals are able to make conclusions about people they have never met before due to the mental shortcuts provided by social categories (Quinn, 2011; Rips, 1975). In the early stages of cognitive development, children utilize abstract and intuitive notions regarding the structure of particular domains to facilitate systematic categorization. For instance, they possess the belief that animals belonging to the same species share common biological characteristics, while artifacts within a similar category share functional attributes (Gopnik & Wellman, 2012; Wellman & Gelman, 1998).

It seems that children develop an early understanding that individuals within a group have distinct responsibilities towards each other, which is a component of their abstract perception of the social environment (Rhodes, 2013a). Infants, by the age of two, possess the cognitive anticipation that individuals within a social group will engage in reciprocal assistance. An exemplary illustration can be found in the research conducted by Jin and Baillargeon (2017), wherein infants at the age of 17 months were subjected to two distinct groups that were distinguished by labels and perceptual attributes. Subsequently, the authors presented scenarios wherein a

representative from each respective group necessitated assistance in attaining a specific objective, and a representative from one of the groups possessed the capability to offer aid. Infants exhibited a heightened level of attention (indicating a deviation from their anticipated outcome) when the agent chose to assist an individual from a different group rather than an individual from their own group. This observation suggests that infants possess a conceptual understanding that members within their group are more inclined to provide assistance to one another compared to individuals from outside groups. Furthermore, it has been observed that infants aged one and two years demonstrate the ability to anticipate agents selectively distributing resources to members of their own group, even in situations where achieving equal distributions is not possible (Bian et al., 2018). Additionally, these infants exhibit a tendency to avoid assisting individuals who have previously caused harm to members of their own group (Ting et al., 2019). Therefore, this implies that the underlying premise that individuals within a shared group tend to cooperate and support each other to a greater extent than individuals from different groups is an aspect of children's comprehension of the social context that emerges at an early stage, and is also a fundamental component of adults' perception of intergroup relations. Additional support for this assumption is derived from research investigating the cognitive processes of older children when reasoning about unfamiliar social categories with which they lack familiarity and do not associate themselves.

According to Rhodes (2012), by the age of three, children demonstrate a tendency to expect members of unfamiliar social groups, which may be intentionally constructed, to harm individuals from other groups rather than individuals from their own group. Rhodes and Chalik (2013) posit that individuals perceive it as morally reprehensible for an agent to cause harm to a person belonging to the same group as the agent. Furthermore, Chalik and Rhodes (2015) and Rhodes (2014) argue that individuals may employ their group identity as a justification for engaging in harmful behaviour towards members of other groups while simultaneously assisting members of their own group. Moreover, it is expected that children in this age group possess the expectation that members of different groups will provide assistance to one another in situations involving intergroup conflict (Chalik & Rhodes, 2014). The aforementioned presumption that individuals within a group will provide assistance to one another in times of discord may also elucidate the reason why infants expect the

formation of larger groups during conflicts between different groups (Pun et al., 2016).

According to various theoretical frameworks, the inclination to anticipate that social groupings will exhibit patterns of cooperation and rivalry is an inherent and essential mechanism for effectively navigating the social environment. This perspective posits that certain aspects, such as group loyalty or an ethic of community, are inherent in individuals' moral inclinations. It is suggested that morality may have originated as a mechanism for regulating behaviour among large groups of unrelated individuals who relied on cooperative efforts. This viewpoint is supported by research conducted by Graham et al. (2011), Rai and Fiske (2011), and Shweder et al. (1987). There have been assertions made regarding the significance of individuals' dependence on group members for survival and procreation, which subsequently contributed to the emergence of innate capacities to discern patterns of allegiance (Kurzban et al., 2001; Pietraszewski et al., 2014). In accordance with this perspective, it is posited that children acquire the ability to recognize and react to social categories that they encounter in their immediate environment, such as gender, social class, and nationality. These categories serve as cues for the formation of social affiliations. However, it should be noted that certain cultural divisions, particularly those based on language cues, may elicit responses from newborns due to their historical significance as reliable markers of social allegiance throughout the course of human development. This notion is supported by research conducted by Cosmides et al. (2003), Ferera et al. (2018), Kinzler et al. (2010), and Spelke & Kinzler (2007).

Moreover, an alternative possibility is that the cognitive frameworks of infants and young children limit their understanding of social categories, thereby influencing their behavioural expectations due to a heightened tendency to perceive these categories as indicative of normative frameworks. As an illustration, the early-developing biological domain representations in children also suggest a proclivity to employ categories in order to govern the expected behaviour of group members. This can be observed in their belief that horses are expected to run quickly, and that a horse deviating from this norm is considered abnormal (Foster-Hanson et al., 2018; Foster-Hanson & Rhodes, 2019a; Haward et al., 2018). From this perspective, the viewpoints of children regarding the obligation of individuals in a group to assist each other, the obligation of animals to display their inherent characteristics, and the obligation of

artifacts to fulfil their intended purposes all indicate a prevailing belief that categories determine the expected behaviours of their members. These viewpoints are influenced by learning processes that engage with specific domain-related information.

1.3.1.3 Social Processes. It's important to note that social motivational processes are just as crucial to this growing system as the perceptual and conceptual mechanisms that support social categorization. In particular, kids are driven to learn about the categories to which they belong, to find others who are "like them" (Meltzoff & Moore, 1977), and to learn about and develop the characteristics shared by their own social groups. According to the social identity theory advocated by Nesdale and Flessner (2001) and Tajfel and Turner (1979), social categories possess a unique characteristic wherein individuals are able to associate themselves with members of these groups. From this perspective, the process of self-identification actively promotes the formation of social categorization. In fact, a substantial body of research centered on the minimal group paradigm (Sherif et al., 1961; Tajfel, 1981; Tajfel & Turner, 1979) contends that an own-group positivity bias develops in response to any aspect of similarity shared by two people, regardless of whether the similarities are cooperative or competitive in nature. The act of classifying an individual as either belonging to an in-group or an out-group has been found to promote social evaluations and the use of inductive reasoning, leading to the amplification of positive characteristics for in-group members and negative characteristics for out-group members (Dunham, 2018; Nesdale & Flessner, 2001; Patterson & Bigler, 2006).

Furthermore, it has been observed that individuals exhibit a favourable bias towards their own social group even in the absence of specific knowledge about the group or any expectations regarding future interactions with its members. This suggests that mere membership in a social group has an impact on the formation of evaluations between different groups. According to Dunham (2018), the utilization of random allocation to a distinct group of children is adequate to elicit social motivations among preschool and elementary school-aged children. These motivations include increased engagement with members of their own group, enhanced trust in their group members, greater willingness to share with their group members, and a more favourable evaluation of their group members. Furthermore, the phenomenon of children's affiliation with their social group leads to a biased

recollection of pro-social and antisocial behaviours exhibited by both in-group and out-group members. This biased recollection serves to strengthen a favourable evaluation of the in-group (Baron & Dunham, 2015; Dunham, 2018).

The origins of these social motivations, which involve the recognition, attachment, and positive evaluation of social groups, appear to emerge during the first year of life. Infants demonstrate a preference for individuals who share similarities with them, such as those who make similar choices (Mahajan & Wynn, 2012) or speak the same dialect (Kinzler et al., 2007; Meltzoff, 2007). Additionally, Pun et al. (2018) found that infants anticipate pro-social behaviour from individuals who speak their dialect, which may explain their inclination to affiliate with, receive resources from, and imitate familiar individuals. It is worth noting that in these studies, familiar individuals often belong to the same linguistic group as the infants.

Together, these results demonstrate that membership to a group serves as a schema for constructing one's self-awareness in a way that supports and sustains a favourable perception of one's own group. Schemas organize knowledge, direct attention, and reinforce ideologies that are core to schema across all domains (Bem, 1981; Carter & Levy, 1988; Piaget & Garcia, 1989; Martin et al., 1995). Conclusively, social motives such as the need to label others as belonging to an in-group or an out-group and an early awareness of intergroup relations drive the basic process of social categorization during childhood.

1.3.2 Psychological Functions of Social Categorization

Social categorization provides a fundamental epistemic function: it organizes and structures our understanding of the social context. In their ability to serve these primary knowledge functions, social categories are no different from other notions. The process of categorizing an individual within a particular social group, based on factors such as demographic characteristics, societal norms, familial connections, shared interests, or other relevant social markers, facilitates the drawing of important and necessary conclusions on a variety of matters. It is possible to make estimations regarding an individual's objectives, intentions, competencies, knowledge, and fundamental personality traits. These findings can be highly beneficial in determining the appropriate approach and manner of engaging with other individuals, much like the classification of physical objects aids in guiding our interactions with

them (for example, we know that "writing " is an appropriate interaction with a "pen").

In one fundamental way, categorizing people differs from categorizing objects. When we assign a social category to a person, we are inclined to evaluate our own position within that grouping (i.e., as a member or non-member). Consequently, social categorization assists us in connecting with others who hold our group memberships (i.e., in-groups); nevertheless, it also has the capacity to construct psychologically significant separating lines between the observer and the recipient (i.e., out-groups). Hence, social categories serve a vital role in shaping the perceiver's sense of identity, fostering a sense of belonging or detachment from others, in addition to their cognitive function of facilitating knowledge acquisition. Tajfel (1969, 1982) established a robust theoretical framework for examining the impacts of epistemic and identification tasks presented by social categories.

Category use is adaptive and useful, if not necessary, for understanding the social world, but it can also have undesirable outcomes. Stereotypes concerning the typical qualities of social groups are often beneficial tools for generating meaningful perceptions of others, rather than the "rotten generalizations that smelled up the mental household" (Schneider, 2004, p. 562) that were presumed in the initial research. However, in order to effectively fulfill their assigned epistemic functions, these generalizations must possess a satisfactory level of accuracy. Nevertheless, the research indicates that the precision of specific stereotypical perspectives can be constrained by various factors. The formation of accurate stereotypes relies on the exposure to appropriate and unbiased representations of individuals within a particular group. Given the prevalence of regular interactions between individuals of different genders, it is not surprising that numerous commonly held gender stereotypes have been empirically validated (Swim, 1994). Furthermore, when one does not have much interaction with members of a specific group, their views about that group are mediated by what others say about that group, and this information can be systematically distorted (Allport & Postman, 1947). Cognitive distortions of a systematic nature may arise when individuals hold strong pre-existing beliefs about a particular social group, leading to biased evaluations of newly encountered members of that group (Cameron & Trope, 2004; Hamilton & Sherman, 1994). Typically, individuals may prioritize their intense psychological needs over concerns regarding

the accuracy of their knowledge, leading them to favour interpretations that satisfy their motivations, even if these interpretations deviate from a realistic comprehension of reality (Kunda, 1990). Hence, it can be argued that generalizations pertaining to social groups serve not only to fulfil personal ego gratification and system justification, but also play a significant role in the acquisition of knowledge. The fulfilment of these motivating goals does not necessitate the presence of accurate beliefs.

1.3.3 Social Categories

A *social group* is comprised of a minimum of two individuals who engage in on-going interactions characterized by shared expectations and a common sense of identity. This description elucidates the fact that individuals are interconnected with diverse social groups, encompassing familial ties, different circles of friendships, social strata, economic clusters, professional environments, affiliations with clubs, and memberships in organizations, among others. With the exception of extraordinary situations, it is implausible to conceive of any individual living in complete solitude. Even individuals who maintain an independent lifestyle still interact with family members, colleagues, and acquaintances, and to a certain degree, participate in diverse social circles.

A *social category* is comprised of individuals who possess at least one common characteristic, although their interactions with one another may not be a necessary condition. The concept of "man" serves as a representation of a social category. Despite the absence of interaction, all males possess a fundamental similarity in the form of their biological sex. Asian Americans constitute a distinct social group, characterized by their shared ethnic heritage and United States citizenship, despite potential variations in communication and other shared attributes. The aforementioned instances demonstrate that gender, race, and ethnicity constitute the fundamental basis for numerous social classifications. Additional commonly recognized social categories encompass religious affiliation, geographical location, and socioeconomic standing. The transformation of a social category into a social group occurs when individuals within the category engage in interactions and mutually acknowledge their membership within the group.

Human groups can be conceptualized as prototypes or complex sets of interconnected traits that individuals cognitively represent, encompassing both shared characteristics within groups and distinctions between groups. Prototypes maximize entitativity (the degree to which a group is unique and well-defined) and minimize meta-contrast (the degree to which maximal similarity among members of the group is balanced with the maximal difference between the group as a whole and a relevant out-group). When we hear "French," our prototype of that national group comes to mind—possibly something to do with berets, baguettes, and bicycles. However, if we compare the French to the Dutch, bicycles may not be part of the prototype because it is not a quality that distinguishes the categories in a sharp way—the Dutch also cycle a lot. Binary categorizations are commonly constructed, wherein one of the categories pertains to the collective to whom individuals belong, referred to as the in-group. Prototypes serve to not only highlight shared characteristics within the in-group, but also underscore the differences between our group and specific out-groups. The prototypes of in-groups may vary based on the out-group with which our group is being compared. Prototypes exhibit context-dependency.

1.3.3.1 Cognitive Representations of Social Categories. It is crucial to have cognitive representations of social groupings in order to (a) identify people who belong to a specific category and (b) make judgments about those people who do. According to the classic theory of categories (Katz, 1972), the existence of a category is determined by a group of criteria that, taken separately and collectively, sufficiently describe the category. Due to numerous conceptual criticisms and conflicting empirical evidence, this viewpoint was widely disregarded, and two competing alternatives were adopted. According to Rosch (1978), the probabilistic viewpoint suggests that categories are defined by a collection of prototypical attributes. Individuals' perceptions of category membership are influenced by the degree of similarity, also known as "family resemblance," between a particular instance and the prototype of the category. Additionally, the exemplar view, as proposed by Medin and Schaffer (1978), presents an alternative perspective suggesting that a category is most accurately represented by the distinguishing attributes of its most prominent individual exemplars rather than relying on a singular fixed category prototype. From an exemplar viewpoint, the category is represented with minimal abstraction; features of specific instances serve as the defining criteria. Sherman (1996) argued that both

perspectives are valid but at distinct stages of the building of group representations. While facing members of a novel group for the first time, exemplar-based representations appear to drive category judgments, whereas probabilistic, prototype-based representations seem to establish after sufficient exposure to group members.

People do, evidently, maintain significant beliefs about the characteristics and features related to social groups, irrespective of the representational framework one assumes. Categories are typically described by their descriptive qualities, but their representations encompass more than a mere enumeration of qualities that are individually linked to category membership. Hence, the aforementioned attributes are encompassed in causal theories that explicate the underlying reasons for the characteristics of the category in question (McGarty et al., 2002; Murphy & Medin, 1985), as opposed to merely providing a descriptive account of the category. Certain features possess a characteristic known as "causal status" (Ahn et al., 2000) as they play a role in the formation of other features within the category, which are referred to as effect features. For example, if a community is commonly perceived as exhibiting traits of diligence, high educational attainment, and prosperity, the characteristic of being hard-working may be considered a prominent element that causally influences the perceiver's cognitive depiction of the group. This perception can help elucidate the community's accomplishments in terms of education and financial well-being. According to Rehder and Hastie (2001), when making evaluations regarding category membership and drawing inductive conclusions about category members, features that possess a causal status are accorded greater significance compared to features that are considered as effects.

When examining social categories, as well as other categories considered being "natural types," it is commonly assumed (either implicitly or explicitly) that the defining characteristics of a category are attributed to an inherent essence (Rothbart & Taylor, 1992; Yzerbyt & Rocher, 2002). The concept of "psychological essentialism" (Medin & Ortony, 1989) emerges during early childhood and encompasses the belief that categories possess an intrinsic and fundamental essence that gives rise to their observable characteristics (Gelman, 2003). According to this perspective, the primary determinant of category judgments is not surface-feature similarity, commonly referred to as "effect" similarity. Instead, the decisive factor is the presence or absence of the category essence.

Psychological essentialism offers an intuitive ontological template for comprehending the natural world that does not require explicit assumptions about what the inner essence embodies, however developments in genetic studies have supplied an apparently comprehensive basis for speculating about the underlying inner cause (or essence) defining category membership: DNA. Now, for many types of social groups, genetic determinism has been associated with psychological essentialism, with genes serving as the ultimate explanation for a group's qualities (Dar-Nimrod & Heine, 2011; Keller, 2005). The fact phenotypes represent the interplay between nature and nurture and that there is much evidence that gene expression is generally influenced by the environment (Gilbert, 2005; Jaenisch & Bird, 2003) raises questions on the scientific validity of this perspective (Bronfenbrenner & Ceci, 1994). Consideration of these types of biological plasticity is lacking in essentialist thinking, resulting in representations of many social groups (e.g., gender and racial groups) are based on implicitly essentialist concepts emphasizing the immutability of group features (Haslam et al., 2000).

The nesting and hierarchy of categories reflect a different facet of social category representation. "Punjabi" is embedded within broader, superordinate categories (such as "Indian origin," "human beings," "carbon-based life forms," and so on). It can then be characterized in terms of increasingly narrow subcategories (such as "Punjabi speaking" or "Belonging to Punjab state"). The categories located at the center of this hierarchy are commonly referred to as "basic" (Rosch et al., 1976) due to their status as the first categories acquired, named, and utilized during infancy. These categories also serve as the level at which a significant portion of world knowledge is organized. While it has been observed that individuals with extensive knowledge in a particular field tend to employ a greater number of subordinate levels within a category (Tanaka & Taylor, 1991), it is at the basic level where the differentiation between categories is most pronounced. This makes the basic level the most advantageous for establishing conceptual distinctions (Markman & Wisniewski, 1997).

Much research in the field of stereotyping has focused on testing the concept that representations of basic-level social categories, which appear to be so beneficial for routine distinction-making, are safeguarded by subtyping (Richards & Hewstone, 2001). Perceivers are inclined to create a specific subcategory that is considered a

particular case, an "exception that proves the rule," when they confront group members who do not demonstrate group-typical features (Kunda & Oleson, 1997). The original stereotype does not need to be altered in this way. Recent theoretical speculation has shown a marked increase in the theoretical vitality of ideas about category representation. While it is evident that there is a significant level of consistency in these representational states, it may be more prudent to conceptualize mental representations as transient states rather than enduring realities.

1.3.3.2 Social Categories in India. Through literature and general observations there are four broad ways in which social categories can be differentiated which were given by Berreman in 1972:

1. Muslim and Sikh religious groups as opposed to Hindu, Christian, Jain, Parsi, and so on.
2. Regional-linguistic-national-racial groups, such as Punjabi and Pahari, as opposed to Hindiwill (person of Hindi speaking area), Madrasi (South Indian, sometimes comprising people from western India: Marathi and Guajarati), Bengali, Anglo-Indian Englishman, Tibetan, Nepali, and so on.
3. Caste categories (the varnas): Brahman, Kshatriya, Vaisya, Shudra; "high castes" (usually the highest three varnas uarnasthe "twice born"); "small castes" (essentially the Shudra varna of low-status artisans); "low castes" (primarily untouchables); and so on.
4. Social class, lifestyle, and vocational classifications: peasant, labourer, well-to-do, white-collar, and so on.

Each of these broad categories contains discrete subsets that are distinct from and often conflicting with one another; it would be impossible to provide an appropriate presentation of all of these complexities in a two-dimensional paradigmatic outline in Indian context as well as category status of subsets and nomenclature changes geographically and culturally.

1.3.4 Multiple and Cross-Categorization

Social categorization describes the intergroup relationship in ways that are not necessarily straightforward and clear-cut but are typically complicated and multi-layered. People can be described in a wide variety of ways, some of which overlap and are intertwined with one another, such as being young, Indian, African, female,

Hindu, or an architect. There are several aspects, such as context and motivation, that influence which identity is utilized when perceiving a person; however, it is essential to keep in mind that these identities are not exclusive of one another. We can and usually are identified, and we identify others based on a number of group identities: disabled men, young Indian citizens, and female doctors. Multiple social categorization research has revealed that our complex web of intertwined and overlapping identities has far-reaching consequences for the study of prejudice and the mitigation of intergroup bias. On some dimensions, every person with whom one interacts will be an in-group member, but on others, they will be an out-group member. Multiple categorization studies have found that individuals may and do utilize multiple bases for social categorization at the same time (Crisp & Hewstone, 2007; Crisp et al., 2001).

One method for examining impression formation in multiple categorization settings is the crossed-categorization paradigm. Cross-categorization research was motivated by social anthropology's discovery that cultures with more people belonging to many groups experience less social conflict (Gluckman, 1956; LeVine & Campbell, 1972). Crossed categorization, as used in the socio-psychological literature, is the process of combining two components of social categorization to create four targets: a double in-group, a double out-group, and two targets who belong to overlapping group memberships (Deschamps & Doise, 1978). For instance, rather than observing simply females against males or young against old, perceivers pay attention to both dimensions in crossed-categorization scenarios as illustrated in table 1.1 below:

Table 1.1

Crossing the Gender and Age Dichotomous Category Resulting in Four Combinations of Social Groups

Young (in-group)		Old (out-group)	
Male (in-group)	Female (out-group)	Male (in-group)	Female (out-group)
Young male (in-group/ in-group)	Young female (in-group/ out-group)	Old male (out-group/ in-group)	Old female (out-group/ out-group)

In this context, individuals of both genders can be regarded as belonging to a comparable group, as they can exhibit characteristics of either youth or old age. Deschamps and Doise (1978) argue that the phenomenon of categorizations intersecting with one another should result in conflicting category differentiation processes. The process of differentiation in a single dimension will be counterbalanced by the process of assimilation in another dimension. Numerous studies have been conducted to investigate explicit evaluations (Crisp et al., 2001; Deschamps & Doise, 1978; Diehl, 1990; Vanbeselaere, 1987, 1991, 1996), implicit evaluations (Crisp et al., 2006b), and assessments of stereotype threat (Rosenthal & Crisp, 2006; Rosenthal et al., 2008). These studies collectively provide support for the hypothesis that evaluations among mixed groups with a shared in-group identity tend to exhibit reduced differentiation and diminished biases.

1.3.5 Patterns of Cross-Categorization

Typically, scholarly investigations commonly demonstrate the absence of intergroup bias in situations involving crossed-categorization, alongside the presence of negative evaluations towards the double out-group targets (Diehl, 1990). Alternatively, bias tends to be diminished in overlapping conditions, resulting in the most positive evaluations of double in-groups, less positive evaluations of partial in-groups (without negative evaluations), and ultimately negative evaluations of the double out-groups (Brown & Turner, 1979; Vanbeselaere, 1987). The pattern in question is commonly known as the *additive pattern*, as identified by Brewer (1968) and Hewstone et al. (1993). The fundamental crossed categorization effect was found to exhibit an additive pattern in two meta-analyses conducted by Migdal et al. (1998) and Urban & Miller (1998). According to Urban and Miller (1998), the baseline pattern is additive, but there exists considerable variability in the patterns that extend beyond it. This phenomenon primarily arises in research studies that employ real-world social categories and can be attributed to the unequal psychological importance of categorization dimensions.

According to Brewer et al. (1987) and Commins & Lockwood (1978), the *category dominance pattern* states that individuals tend to focus solely on the primary categorization dimension while disregarding the secondary, subordinate dimension. The category conjunction pattern, as identified by Rogers et al. (1981), encompasses two distinct variants: *conjunction similarity*, which pertains to *social inclusion*, and

conjunction dissimilarity, which pertains to *social exclusion*. In the context of the conjunction similarity pattern, targets that possess at least one shared group membership with the participant are accorded equal favourability as the targets that possess both group memberships. According to the dissimilarity pattern in conjunction, targets that differ from the participant on at least one dimension are assessed with the same level of negativity as targets that are distinct from the respondents on both dimensions.

The *hierarchical ordering pattern* proposes that the utilization of the second category is contingent upon whether the individual in question has been classified as an in-group or an out-group member on the initial dimension (referred to as the "first" category, which is considered more significant). According to Brewer et al. (1987) and Park & Rothbart (1982), the *hierarchical acceptance pattern* indicates that only targets classified as in-group members on the initial, significant dimension will undergo further differentiation based on the second dimension. Conversely, targets classified as out-group members on the significant dimension will not undergo further differentiation based on the second dimension. According to Hewstone et al. (1993), the *hierarchical rejection pattern* entails that differentiation on the second dimension, which is of lesser importance, will only take place for targets that have been classified as out-group members on the first dimension.

Ultimately, it is also possible that no differentiation takes place across any of the dimensions, resulting in what is referred to as the *equivalence pattern*. A comprehensive compilation of the various patterns identified in existing scholarly works can be found in Table 1.2. These patterns were exclusively proposed and tested using a two-dimensional categorization framework. Numerous studies in the field of categorization primarily center their attention on examining the potential of multiple categorizations in reducing bias (Hall & Crisp, 2005; Prati et al., 2016, 2015). The aforementioned patterns possess a high degree of informativeness in relation to comprehending the processes of impression formation within various categorization contexts. Nevertheless, the issue regarding the reasons behind the emergence of dominant categorization dimensions has not been adequately explored. According to Hagendoorn and Kleinpenning (1991), the significance of particular qualities is contingent upon the unique societal context, group categorization, and historical timeframe. According to Brown (2010), the determination of the dominant category

dimension in a given situation is contingent upon specific local factors (p. 47). In this study, we exclusively employed categories that were pertinent to the specific social context at the particular time. This approach also facilitates the examination of the psychological foundations of category dominance.

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Table 1.2*Evaluation Patterns Under Crossed Categorization Situation*

Conditions	Pattern	Dimension 1: In-group		Dimension 1: Out-group	
		Dimension 2: In-group	Dimension 2: Out-group	Dimension 2: In-group	Dimension 2: Out-group
Baseline	1. Additive	2	0	0	-2
		ii	>	oi	>
Differential importance (here dimension 1 is more important)	2. Dominance	1	1	-1	-1
		ii	=	oi	=
Positive affect/ in-group prime	3. Social inclusion	1	1	1	-3
		ii	=	oi	>
Negative affect/ out-group prime	4. Social exclusion	3	-1	-1	-1
		ii	>	oi	=
Positive affect/ differential importance (here dimension 1 is more important)	5. Hierarchical acceptance	4	0	-2	-2
		ii	>	oi	=
Negative affect/ differential importance (here dimension 1 is more important)	6. Hierarchical rejection	2	2	0	-4
		ii	=	oi	>
Extreme positive affect/low importance groups/absence of salience categorization	7. Equivalence	ii	=	oi	=

Note. Adapted from (Crisp & Hewstone, 2007). Abbreviations: i, in-group constituent of group; o, out-group constituent of group. The presence of positive or negative signs indicates the locations where variations can be anticipated among the four crossing subgroups of categories. Contrast weights cannot be demonstrated for the equivalence pattern due to its null effect.

1.4 Intergroup Relationship

Individuals are social creatures who constantly seek out and rely on their fellow group members. All of an individual's actions, from their worldview to their emotional state to their behaviour, are influenced and, at times, constrained by the group(s) to which they belong. There are many different kinds of social groups that someone can be a part of, and they are often defined by characteristics such as a person's gender, race, occupation, religion, language, etc. These groups may be inclusive to varying degrees. For instance, a single ethnic group will be composed of males and females, each of whom will be a member of a distinct set of occupational and social groupings. As a result, there are higher-order group memberships that include less inclusive social groups. Therefore, social psychologists have arrived at the view that the relationships between the multiple groups that make up any particular social system are characterized by a small number of distal structural components (Fiske et al., 1999).

It is necessary to begin with a working definition of "group" in order to proceed with developing a working definition of "intergroup relations." This phenomenon represents particular challenges due to the diverse range of conceptualizations of a group within the field of social psychology, which corresponds to the varying interests of researchers investigating this topic. For example, certain scholars argue that a group can be defined as a limited assembly of individuals who interact in a manner that fosters interdependence (Arrow et al., 2000). There are differing viewpoints among researchers regarding the concept of a group. Tajfel and Turner (1986) propose that a group emerges when individuals define themselves based on the unique characteristics of a social category that includes them. On the other hand, Allport (1924) argues that the notion of a group is a logical error, asserting that any psychology pertaining to groups is fundamentally and entirely a psychology of individuals. The presence of divergent interpretations often stems from underlying metatheoretical conflicts regarding the extent and objectives of the social psychology field pertaining to group dynamics and intergroup relations (Abrams & Hogg, 2004).

Considering that social categorization is probably a cognitive process, the definition of the social group that rest on the foundations of cognitive processes and forms the basis of social identity theory is being used as an operational definition in

the present research (Abrams & Hogg, 2010; Hogg, 2006, 2012a). Psychologically, a *group* emerges whenever two or more individuals share a shared self-inclusive category within which they describe and evaluate themselves. But, it is essential to recognize that existence in a group entails more than simply defining oneself; it entails social contact, interdependent desires and sentiments, and so on and so forth. Sherif's (1962) classic and generally commonly recognized definition of intergroup relations has been adopted as an operational definition of intergroup relationships in this study since it is consistent with cognitive notion of social groups. According to Sherif (1962), "Intergroup relations are the relationships between two or more groups and their members. Intergroup conduct occurs whenever members of one group interact, collectively or individually, with members of another group on the basis of their group identifications" (p. 5).

1.4.1 Determinants of Intergroup Relationship

There are two aspects of society that account for the nature of interpersonal relationships. The relative dominance or subordination of one group over another within a given social system is a primary factor in determining the nature of intergroup relations. The second characteristic concerns how groups interact with one another, namely whether or not they are benevolent (cooperative) or malevolent (competitive). Both of these elements have been shown in countless studies to have a major impact on how interactions between groups develop over time (Yzerbyt & Demoulin, 2010). Our thoughts and views, i.e. our prejudices about the different groups in our societies are shaped by a number of factors, including the groups' position and perceived competition. The Stereotype Content Model (SCM, Fiske et al., 1999, 2002) proposed that social judgments about groups fluctuate systematically along two dimensions: warmth (or friendliness) and competence (Asch, 1946; Rosenberg et al., 1968; Wojciszke, 2005). While warmth judgments indicate a group's good or bad intentions towards us, competence judgments indicate the group's ability to carry out those intentions. Competence evaluations are predicted by one's social standing inside a group, while warmth ratings are mostly determined by one's participation in group competition. A group's perceived competence and warmth will both increase with its standing and decrease with its perceived rivalry (Cuddy et al., 2008). It has been demonstrated that the strength of these two social structural elements in determining stereotypes of warmth and competence is universal, applying

to a wide variety of groups varying from national to professional groupings and being constant across cultural contexts (Cuddy et al., 2009).

1.4.2 Theories of Intergroup Relationship

1.4.2.1 Social Dominance Theory (SDT). Social dominance theory is one of the most well-known frameworks for understanding interactions between social groups (SDT; Sidanius & Pratto, 1999). It tries to comprehend the processes through which hierarchical social relations are established and sustained. SDT approaches intergroup relationships from both a group and an individual perspective. It takes a group-based approach assuming that people belong to three different social groups based on age, gender, and an "arbitrary set system" that can include things like people's religion, race, or social class (Pratto et al., 2006). These groups have different amounts of social power and material resources.

According to the SDT, the experience of intergroup relationships differs not just between the dominant and subordinate groups but also among the three systemic groups. As a result, intergroup bias is not uniform among social groupings. For instance, high-status groups have a greater propensity to exhibit in-group favouritism, whereas low-status groups exhibit this tendency to a lower extent. A term for this phenomenon is "behavioural asymmetry" (Sidanius & Pratto, 1999, p. 227). That is to say, compared to members of low-status groups, high-status group members are more likely to engage in intergroup behaviours that benefit them. The acceptance of Legitimizing Myths (LMs) in order to achieve behavioural asymmetry is a fundamental concept of the Social Dominance Theory (SDT). They may either be Hierarchy Enhancing or Hierarchy Attenuating (HE-LM and HA-LM, respectively). LMs are any set of ideas, beliefs, norms, or values that advocates for either the upkeep of existing social inequalities between different groups (HE-LMs) or for greater social equality (HA-LMs) (Sidanius & Pratto, 1999). Some of the most well-known HE-LMs include the protestant work ethic, belief in a just universe, and prejudice and sexism. Examples of HA-LMs include things like feminism, socialism, and universal human rights, among others.

Within the framework of Social Dominance Theory (SDT), the concept of social dominance orientation (SDO; Pratto et al., 1994) serves as a core construct. It evaluates individual differences in their desire for group-based domination and

inequalities. SDT is primarily investigated using this theoretical instrument, which provides the theory with an individual approach to the analysis of intergroup relations. As an illustration, SDT suggests that ideologies that favour social hierarchy and discourage social equality are to blame for a wide range of discriminatory practices. These ideological values vary not only between individuals but also between groups and can be anticipated based on the levels of SDO. From these considerations, we can hypothesize that individuals with higher SDO levels exhibit more HE-LMs, while those with lower SDO levels exhibit more HALMs. This hypothesis has been supported by studies showing that those in positions of social dominance are more likely to advocate for HE-LMs than those in positions of low dominance (Quist & Resendez, 2002; Sidanius et al., 1994). In addition, researchers have discovered that HE-LMs have a favourable correlation with group attachment among those who belong to high-status groups. Members of lower-status groups report a weaker or even negative correlation. In contrast, among high-status groups, HA-LMs have been reported to be adversely related to group attachment. Members of lower-status groups experience a weaker or even reversed version of this association (Levin et al., 1998). As a group, men have a higher SDO than women do, and persons who are members of dominant groups, in general, have a higher SDO than those who are members of subordinate organizations (Pratto et al., 2000; Sidanius et al., 2000). Gender identity plays a role in the way men's and women's SDOs differ (Dambrun et al., 2004; Wilson & Liu, 2003). In addition, individuals employed in occupations that attenuate hierarchy tend to exhibit lower levels of social dominance orientation (SDO) compared to those in roles that enhance hierarchy (Sidanius et al., 2004). Furthermore, it is worth noting that the discrepancy in social dominance orientation (SDO) among groups that have been arbitrarily formed becomes more pronounced as the status and power differential between these groups increases (Levin, 2004; Sidanius et al., 2000). The collective evidence suggests that the concept of social dominance orientation (SDO) is not a fixed trait, but rather influenced by contextual factors.

Pratto et al. (2006) suggested the idea that five overarching forces affect SDO in order to explain the heterogeneity in SDO seen both between and within groups. In addition to factors such as group status, gender, and the social setting, as was previously demonstrated, these factors also included personality and socialization. It

has been shown empirically that SDO is inversely associated with traits including openness, agreeableness, and sensitivity (Akrami & Ekehammar, 2006; Duckitt et al., 2002). Empirical research conducted by Akrami and Ekehammar (2006) and Duckitt and colleagues (2002) has demonstrated that social dominance orientation (SDO) has a negative correlation with personality qualities such as openness, agreeableness, and sensitivity. According to Pratto et al. (2006), the inventors of social dominance theory (SDT) asserted that socialization into religious beliefs, as well as highly traumatic experiences, intercultural contacts, and education, are all examples of socializing influences that have the potential to alter individuals' degrees of SDO.

1.4.2.2 Self-Categorization Theory (SCT) and Social Identity Theory (SIT). Social identity theory is among the most comprehensive theories of intergroup relations (SIT; Tajfel, 1982b; Tajfel & Turner, 1979). SIT is more interested in intergroup relations (acting in terms of the group) and social identity than in self-identification or interpersonal behaviour. It provides a group-level investigation of intergroup relations and bias. The structure of SIT recognizes social conduct as existing along a continuum that encompasses both interpersonal and intergroup dynamics. In accordance with Turner and Reynolds (2004), individuals exhibit behaviour that is either self-oriented or group-oriented. Nevertheless, it is improbable that either of these behaviours will be observed in their most unadulterated state in real-world situations. It's not likely that people's actions are either wholly determined by or wholly independent of the groups to which they belong. According to empirical evidence, it has been observed that there exists a connection between the self and the in-group (Smith & Henry, 1996). This finding suggests that social behaviour can be understood as a manifestation of both individuality and the individual's identification with a particular group.

People's social identities are generally formed as a result of their group memberships. According to the Social Identity Theory (SIT), individuals tend to exhibit group-oriented behaviour rather than emphasizing their individual traits in significant intergroup contexts where group membership is prominent. One fundamental assumption in SIT is that social comparisons across groups promote positive in-group distinctiveness. As a result, positive social identity has a favourable impact on personal identity (Turner & Reynolds, 2004). One of the postulates of the Social Identity Theory (SIT) is that people have a tendency to favour their in-group

above other groups, sometimes known as showing in-group favouritism. This tendency was established through empirical research. The initial investigations employing the minimal group paradigm, as conducted by Tajfel et al. (1971), unveiled a widespread inclination towards favouring one's own group, even in situations where there was no apparent conflict between groups. The reason is that indulging in group discrimination makes members of that group feel good about themselves (Lemyre & Smith, 1985). That is, a positive social identity increases their self-esteem.

People's self-perceptions dictate whether they operate on an interpersonal or intergroup level in their interactions with other people. According to Turner and Reynolds (2004), the way individuals view themselves can range from perceiving oneself as an individual (personal identity) to perceiving oneself as a part of a group (social identity). Humans move along a continuum that extends from the interpersonal to the intergroup level, and depending on the circumstances, they categorize themselves individually or collectively. When social identity, rather than one's own personal identity, takes centre stage, it opens the door to the possibility of group behaviour. The aforementioned statement serves as the fundamental basis of self-categorization theory (SCT; Turner, 1999; Turner et al., 1987). The self-categorization theory proposes that variations in self-categorization result in a fluctuating self-perception that encompasses both personal identity and social identity. The self becomes depersonalized when social self-categorization becomes more prevalent; this is because individual qualities are given a lesser degree of significance in the context of intergroup relations.

The Self-categorization Theory (SCT) aims to provide a comprehensive understanding of the factors influencing the categorization of scenarios as either interpersonal or intergroup. This is achieved by examining individuals' self-perceptions. In contrast, the Social Identity Theory (SIT) establishes identity formation along the interpersonal-intergroup continuum. Within the theoretical framework presented, the concept of identity is understood as a contextual and mutable phenomenon, as opposed to a fixed and unchanging entity. It is noteworthy that individuals tend to engage in self-categorization, which in turn leads to the manifestation of self-stereotyping. Similar to how social categorization can result in the formation of stereotypes towards individuals belonging to out-groups, self-

categorization can also play a role in the development of stereotypical perceptions towards oneself.

1.4.2.3 System Justification Theory (SJT). The system justification theory (Jost & Banaji, 1994) is an additional explanation of intergroup relationships. This theory presents a contrasting viewpoint to the notion that in-group favouritism is not a widespread occurrence and that individuals in subordinate positions are not as inclined to endorse in-group favouritism as those in dominant positions (Rudman et al., 2002). Conversely, individuals assimilate perspectives that contribute to the preservation of existing societal norms, even if such adherence may prove detrimental to the interests and well-being of their respective communities. By exposing the situations in which individuals within a group may adopt positions or practices that are at odds with the interests of their group, SJT provides a novel perspective on the study of intergroup dynamics.

According to SJT, the dominant often embraces the ideals of the dominant (Jost & Banaji, 1994). In this setting, the System Justification Theory reintroduces an out-dated Marxist notion called false consciousness, which is claimed to be essential when trying to provide evidence for system justification (Jost & Banaji, 1994). False consciousness arises when people from marginalized communities participate in the intellectual legitimization of the system that denies them rights to which they are legally entitled. False consciousness can be avoided by avoiding participation in the ideological legitimization of oppressive systems (Jost et al., 2001). Jost and Banaji (1994) claim that system-justification beliefs may be attributed to several factors, including an absence of class consciousness, the estrangement of marginalized groups from one another, and diminished levels of identification with the dominant social group.

According to Jost et al. (2003), when individuals attempt to rationalize a social structure that is detrimental to their own well-being or the well-being of their social group, they experience psychological conflicts that involve the intention to justify their own ego and group. The System Justification Theory (SJT) is informed by dissonance theory, which posits that individuals strive to alleviate cognitive dissonance by upholding the legitimacy of the social system, thereby preserving their positive perception of said system. This is done so that SJT has an explanation for the conflicts that have emerged. This is achieved through the utilization of negative

assertions pertaining to either the individual or the collective to whom one belongs (Jost & Banaji, 1994; Jost & Hunyady, 2002). Individuals who face pre-existing disadvantages are more prone to engaging in system justification, even when it entails incurring costs, due to the psychological comfort provided by ideologies that endorse the existing system (Jost & Hunyady, 2002). Put simply, when the system is granted increased legitimacy, individuals experience a heightened sense of satisfaction with the overall status quo and their own position within it. Previous studies have provided empirical evidence that lends support to the notion of system justification across various ideological perspectives (Jost & Burgess, 2000; Jost & Thompson, 2000; Sidanius & Pratto, 1999).

1.4.2.4 Realistic Group Conflict Theory. Sherif (1958) introduced a theoretical framework known as the realistic conflict or interdependence theory, which focuses on understanding intergroup relations. The underlying assumption of this perspective is that human behaviour is influenced by individuals' goals and their perceptions of their interdependence in attaining these goals. When two distinct groups possess a shared objective, such as the attainment of prosperity, but this objective is organized in a manner where one group's advancement can only be achieved by causing a detriment to the other group, leading to a relationship where the goals are mutually exclusive and the groups experience negative interdependence, it is anticipated that the intergroup relations will be marked by competition and disharmony. When two groups share a common objective that necessitates their collaboration for its attainment, the resulting intergroup relations tend to be characterized by cooperation and harmony. This is due to the presence of a non-zero-sum goal relationship, wherein the achievement of a superordinate goal requires positive interdependence between the groups. At the interpersonal level, the presence of mutually exclusive goals gives rise to interpersonal conflict and can result in the dissolution of groups. Conversely, the existence of superordinate goals fosters interpersonal harmony, facilitates the formation of groups, and enhances group cohesion. The notion in question was subjected to examination by Sherif and his colleagues through a seminal sequence of field experiments conducted at boys' camps within the United States (Sherif et al., 1961; Sherif, 1966).

In the conducted studies, commonly known as the Robbers Cave experiment due to their implementation at Robbers Cave State Park in Oklahoma, Sherif

deliberately manipulated the goal relations among individuals and groups. As a result, he successfully established cohesive groups, fostered intergroup conflict and hostility, and to a certain degree, facilitated intergroup harmony. In a more specific context, a phase of group formation occurred during which the boys were assigned to two distinct cabins that maintained no interaction between them. Every individual cabin participated in distinct and collaborative camp activities that were specifically designed to foster group cohesion, establish norms, and create structure. In the subsequent phase, the two factions, now self-identified as "The Rattlers" and "The Eagles," participated in a series of structured competitions, such as a tug-of-war, wherein they vied for prizes of significant worth. The phenomenon of intergroup behaviour exhibited a heightened level of extremity. Negative attitudes towards out-groups emerged, accompanied by the use of derogatory language and a noticeable preference for the in-group. In the concluding phase, Sherif devised scenarios that necessitated the collaboration of both groups in order to attain a shared objective that was unattainable individually, such as the collective effort required towing a significantly burdensome malfunctioning camp truck back to the campsite. Through a series of diverse cooperative endeavours, intergroup relations experienced a gradual amelioration. Sherif documents that during the journey back, the boys harmoniously sang together while aboard the bus, and a few individuals from opposing groups even exchanged contact information. Other researchers have utilized different versions of the boys' camp paradigm (Fisher, 1990). As an illustration, a study conducted by Blake and Mouton (1961) involved a cohort of over 1,000 business executives participating in management training programs lasting for a duration of two weeks. Additionally, Sherif's studies have been successfully replicated in various cultural contexts by other researchers (Andreeva, 1984). The findings from these replications demonstrated congruence with the initial Robbers Cave study. Realistic conflict theory aligns with sociological analyses of social behaviour due to its top-down approach in examining intergroup relations. It is not surprising that a substantial amount of scholarly literature has been dedicated to investigating the correlation between ethnic and racial dynamics, the perception of intergroup threat, and the competition for scarce resources (Bobo, 1999; Bobo & Hutchings, 1996).

1.4.3 Social Categorization Perspective in Intergroup Situation

In 2015, Leonardelli and Toh put forward a conceptual framework for intergroup psychology, in which they distinguished among three types of self-categorization when thinking about the self in an intergroup setting, where people place themselves on one side of a dividing line and attribute a distinguishing feature to themselves that is absent from the other side. The researchers have proposed that individuals may experience one of three distinct forms of self-categorization, contingent upon whether the observer's characteristics, the characteristics of another individual, or both are perceived to demonstrate meta-contrast.

1. Intergroup categorization, also known as "*ingroup-outgroup*" categorization, occurs when a person's own characteristics and the characteristics of those outside of their social group are compared and contrasted. As Dovidio and Gaertner (2010) have already pointed out, this perception is analogous to an "us vs. them" mind set. In this case, social categorization leads to in-group and out-group categories. The research has typically concentrated on categorizations involving two groups, even though functionally intergroup categorization could broaden to situations with multiple out-groups.
2. Limiting one's classification to one's own in-group (hence the term "*in-group*"): The feature of the perceiver is subject to meta-contrast, but the features of other people are not. In other words, the perceiver recognizes similarities between himself or herself and the other people he or she encounters along some dimension of interest (thereby perceiving an in-group category) but fails to recognize people who are not similar to the perceiver as belonging to a clear or salient group. If the psychological significance of in-group-out-group categorization is described as "us versus them", then the significance of in-group-only classification could be described as "us". Consider categorizing "my family" as an example for this situation. They speculated that most people would view their family as an in-group with no clear, comparable out-group, despite the fact that such classifications could occasionally be deemed intergroup categorizations.
3. Categorization exclusively predicated on the out-group, hence the designation "*out-group*," The phenomenon of meta-contrast is observed when there is a contrast between the characteristics of others, but not between the characteristics of the perceiver. Individuals who are perceived as different from the observer are

often categorized as an out-group, comprising individuals who possess certain similarities. Individuals who do not belong to this particular group are not considered to possess sufficient similarity to constitute a distinguishable or prominent category. They used the pronoun "they" to emphasize the psychological significance of the exclusive out-group category. All the three kinds of categorizations are illustrated in table 1.3:

Table 1.3

Forms of Self-Categorization in the Context of Intergroup Relationship

Perceivers' feature(s)	Others' feature(s)	
	Meta-contrast	No meta-contrast
Meta-contrast	Intergroup (us and them)	In-group only (us)
No meta-contrast	Out-group only (them)	No categorization

Note. Table 1.3 showing the kinds of self-categorization in the context of intergroup relationship: Shows how meta-contrast can lead to the development of new categories. Perceivers' belief that individuals share the feature(s) and the feature(s)' perception as different from others leads to categorization, as indicated by the meta-contrast. Individuals' feature(s), another's feature(s), both, or none, could be the focus of an observer's category(s) (source: Leonardelli & Toh, 2015).

The researchers discovered that there were no discrepancies observed among the three different forms of self-categorization and the preconceived notions they had examined concerning self-categorization and the development of social categories. Initially, the emergence of these three self-categorization types can be attributed to the phenomenon of meta-contrast. Furthermore, the authors have denoted these as self-categorizations due to their self-referential nature, wherein they pertain to social categorizations involving in-groups and out-groups. Based on a rigorous interpretation of Turner's (1987) conceptualization of self-categorization, which posits that individuals mentally group themselves with others who share similar characteristics, it can be argued that the establishment of self-categorization necessitates the formation of an in-group category. Consequently, the exclusive categorization of individuals into an out-group category is precluded. Self-categorization theory, as described by Turner (1987), however, does permit the development of a single category, encompassing out-group-only categorizations. Additionally, they contended that out-group categorizations are self-categorizations because they are self-referential (an out-group is an out-group because the perceiver is not a part of it). Third, an "intermediate" level of abstraction is described by these three varieties of self-classification (Turner, 1987). Some may wonder if all three self-categorizations are psychologically real intergroup situations. Understanding how people's views of

groups change as a result of categorization goes beyond simple recognition of the existence of other people to become an important consideration at this point. Categorization aids in perceptual identification (i.e., it provides definition, explanation, or cognitive association by elucidating the factors that cause an object to be compared favourably with some stimuli and unfavourably with others; McGarty, 1999).

1.5 Social Categorization and Intergroup Relationship: Trends and Empirical Investigations

Social categorization and intergroup studies are interconnected fields of inquiry within the realm of social psychology that investigate the cognitive processes and behavioural dynamics underlying individuals' perceptions and interactions with others, contingent upon their membership in distinct social groups. These concepts facilitate our comprehension of the mechanisms through which individuals develop perceptions, attitudes, and behaviours toward individuals belonging to diverse social groups. Because social categorization and intergroup relationships cannot be studied directly, we reviewed the research trends and associated variables used in the context of Social categorization and intergroup studies, Social categorization studies, and intergroup behaviour studies to identify the study aims. Grigoryan et al. (2022) examined the extent to which identification, perceived conflict, and perceived symbolic threat can predict the intensity of intergroup bias across different aspects of social categorization in multiple categorization contexts. A factorial survey experiment was conducted, wherein nine dimensions of social categorization were manipulated in diverse samples from four countries. The experiment consisted of a total of 12,810 observations, with 1,281 participants representing 103 social groups. The dimensions under investigation encompassed age, gender, ethnicity, religion, place of residence, education, occupation, income, and a country-specific dimension, namely caste. The findings of the study revealed that the presence of identification was associated with a greater tendency towards bias, particularly in situations where the participant held a higher social status and the target belonged to a lower social status group. The study found that there was a stronger bias observed when the target of the symbolic threat was a member of a minority group. The occurrence of bias resulting from conflict was found to be limited to a small number of cases. Furthermore, the magnitude and direction of these effects exhibited variability across

countries, dimensions, as well as target and participant groups. The findings presented in this study contribute to our comprehension of the boundaries of generalizability pertaining to established factors that influence intergroup bias. Additionally, these findings emphasize the necessity for developing novel explanations of social-cognitive processes within minority group populations.

Shi et al. (2022) conducted research to investigate how the perceived severity of COVID-19 leads to an increase in biased social distance. In this study, a total of 437 American participants were recruited from MTurk, with ages ranging from 20 to 78 years (median_{age} = 38.50). Thirteen participants were excluded from the analyses due to their failure on one or more categorical manipulation checks. Each participant was assigned randomly to one condition of a between-subjects factorial design with three factors: relative threat level (lower/higher), a priori favourability toward target out-group (less favourable/more favourable), and relevance of threat to target out-group (relevant/irrelevant). The researchers then measured various variables for each participant. The findings of the study suggest that the presence of COVID-19 may lead to an increase in the expression of prejudice, driven by the underlying motive of avoiding disease. Furthermore, the findings of the study indicate that the Bogardus Social Distance Scale provided evidence in favour of the disease avoidance hypothesis. Specifically, it was observed that individuals exhibited a stronger inclination towards social distancing when faced with higher levels of perceived threats compared to situations with lower levels of relevant threats. The findings from a customized COVID-19 Social Distance Scale indicated that participants exhibited a pre-existing bias, as evidenced by their greater inclination to practice social distancing towards individuals of Chinese descent compared to those of Italian descent.

The study conducted by Jha and Khan (2023) aimed to examine gender-based disparities in relation to a range of discriminatory experiences. A sample of 120 working adults, consisting of 60 males and 60 females, was chosen for the study. The participants' ages ranged from 18 to 56, with a mean age of 25.13 ($SD = 11.52$). The study was conducted in the metropolitan city of Delhi, India. The participants were chosen using a maximum variation purposive sampling technique. The researchers evaluated the discriminatory experiences of the participants by employing the everyday discrimination scale. The findings indicated that both males and females encountered various types of discriminatory actions similarly. However, there were

differences observed in the specific scenarios in which males and females experienced discrimination. The researchers have reached the conclusion that in urban area there has been a reduction in gender-based disparities in the experience of discriminatory actions.

In their study, Hackel et al. (2022) employed a series of four experiments to investigate the mechanisms underlying the development of group-based interaction tendencies as a result of interactions with individual members within a group. The study consisted of four experiments wherein participants engaged in a reinforcement learning task that involved a money sharing game. The objective was to examine how participants formed instrumental reward associations with individual group members through direct interaction and feedback. The findings of the study demonstrated that reward learning at the individual level extended to a collective representation, as evidenced by self-reported group attitudes, trait impressions, and the inclination to engage in future interactions with unfamiliar group members. Experiments 3 and 4 provided additional evidence to support the notion that the influence of group-based rewards on interaction choices remained intact even when the previous value of group rewards no longer served as a reliable predictor of future positive outcomes. This finding aligns with the idea that group bias is expressed in a manner akin to habitual behaviour. The findings of this study illustrate a new mechanism for the development of prejudice, which involves the acquisition of biased attitudes through instrumental reward learning during direct interactions with members of specific social groups.

In their research, Long et al. (2022) undertook three separate investigations in order to investigate the impact of intergenerational contact on the perspective-taking abilities of younger adults toward older adults. Additionally, the researchers sought to examine the potential influence of stereotyping and intergroup anxiety in this context. In Study 1, a cohort of 192 college students was administered the Perspective Taking Scale subsequent to being exposed to age-based intergroup relationships as a priming technique. The findings of the study revealed that younger individuals encountered more significant challenges in adopting the perspective of older adults compared to their peers. In Study 2, a total of 200 college students were administered the Prior Contact Scale, Intergroup Anxiety Scale, Negative Stereotype Scale, and Perspective-taking Scale. The findings of the study indicate that there was a positive impact on perspective-taking towards older adults as a result of intergenerational contact. This

effect was observed due to the disruption of negative stereotypes. Additionally, the relationship between intergenerational contact and perspective-taking was influenced by intergroup anxiety. In Study 3, a cohort of 215 college students participated in the completion of the perspective-taking Scale, specifically within the framework of envisioning intergenerational contact. It is noteworthy that the utilization of imagined contact has proven to be an effective method in enhancing the younger generation's capacity to empathize with and understand the perspectives of older individuals. The current study provides empirical evidence supporting the significance of interpersonal contact in shaping the emotional (specifically, intergroup anxiety) and attitudinal (specifically, stereotyping) aspects that play a crucial role in enhancing younger adults' capacity to adopt the perspectives of older individuals.

The study conducted by Abrams et al. (2006) involved an experimental design to investigate the impact of intergenerational contact and stereotype threat on cognitive performance, anxiety levels, intergroup bias, and identification among older individuals. The sample consisted of 32 male and 65 female retired adults, with ages ranging from 59 to 89 years. The M_{age} of the participants was 74.81 years ($SD = 7.43$). The pre-retirement occupations of the participants exhibited a range of diversity, with 12% having worked in unskilled positions, 44% in semi-skilled positions, 21% in skilled positions, and 15% in professional positions. The individuals were enlisted from diverse social groups and organizations catering to retired individuals, specifically within the geographical region of Kent, located in the United Kingdom. The participants underwent individual testing sessions conducted by a female experimenter who was 24 years old. These sessions took place in a secluded room. The session commenced with an examination of individuals' attitudes, experiences, and competencies. Various measures were evaluated, and the participants were instructed to perform a series of cognitive tasks while experiencing either high or low stereotype threat, which was induced by comparing them to younger individuals. Consistent with the theory of stereotype threat, the presence of threat led to a decline in performance. However, this phenomenon did not manifest itself in cases where previous intergenerational interactions had been characterized by more favourable experiences. The impact of contact was moderated by test-related anxiety, as evidenced by the mediating role it played. Consistent with the principles of intergroup contact theory, a correlation was observed between increased positive contact and

decreased levels of prejudice and in-group identification. Nevertheless, this phenomenon was observed exclusively in the high threat condition, as opposed to the low threat condition. The results indicated that the presence of positive intergenerational contact may serve as a mitigating factor in reducing the susceptibility of older individuals to stereotype threat.

The objective of the study conducted by Sivrikova et al. (2019) was to ascertain the relationship between the perception of intergenerational relations and the specific type of generational identification. The study involved a sample of 146 individuals, consisting of 66 males and 80 females, ranging in age from 17 to 72 years. Statistically significant disparities were observed in the perception of intergenerational relationships across various generations, including the post-war, Soviet, transitional, and post-Soviet generations. These disparities were measured using scales that assessed factors such as proximity, mutual understanding, comfort, lack of conflict, and tranquillity. Based on the acquired data, it was deduced that the Soviet and post-Soviet generations exhibit the least favourable relationships. The society demonstrates the most contentious attitude towards individuals belonging to the Soviet generation. The highest level of miscommunication between generations is observed in interactions involving the post-Soviet generation. Those who associate themselves with the transitional generation tend to perceive more negative relationships within their own group compared to interactions outside of the group. Conversely, the most favourable relationships are observed between the older (post-war and Soviet) and younger (transitional and post-Soviet) generations.

According to Van Rossem (2018), the social identity perspective was found to be consistent with generational identity theory. This study utilized a cognitive mapping technique known as the repertory grid method, employing mixed methods. The aim was to investigate the perspectives of Belgian managers from three different generations (Baby Boomers, Generation X, and Generation Y) regarding their own generation as well as the other two generations. The findings of his study indicate those individuals' perceptions of their own generation and other generations can influence the process of social categorization and the development of generational stereotypes. Furthermore, it was observed that some of these stereotypes can manifest in the form of actions or behaviours, and it was also noted that generational stereotypes may not always align with stereotypes based solely on age. A number of

meta-patterns within the stereotypes were also identified. Therefore, his research made a significant contribution to the developing field of study that advocates for an identity-based perspective rather than a cohort-based perspective on generations. Additionally, his work provided validation for the assertion that generations, as a phenomenon in the workplace, should be recognized as a legitimate subject of study.

In their study, Hogg (2023) provided an overview of the analysis conducted by uncertainty-identity theory regarding the motivational aspect of self-uncertainty in relation to group identification and group/intergroup behaviour. The study specifically examined the manner in which self-uncertainty can serve as a motivating factor for individuals to strongly identify with "extremist" groups and identities. The individual emphasized the importance of individuals possessing a distinct and comprehensive understanding of their own identity in order to effectively comprehend the world and their position within it. Additionally, this self-awareness is crucial for strategic decision-making and the ability to predict and interpret the actions of others. Individuals endeavour to minimize their self-uncertainty by aligning themselves with distinct social groups characterized by clearly defined social identities. The occurrence of significant social change has the potential to generate feelings of identity threat, self-uncertainty, as well as alienation and marginalization within society. In order to mitigate personal uncertainty, individuals actively aligned themselves with ethnocentric, populist, and xenophobic factions characterized by a lack of tolerance for opposing viewpoints and the presence of autocratic leadership. This resulted in the creation of distinct identity groups and the reinforcement of divisions between highly polarized factions. In light of the prevailing conditions, the task of dismantling barriers in order to foster reconciliation among polarized and antagonistic factions and identities posed a significant challenge. The speaker concluded the discussion by providing an overview of intergroup leadership theory and its significance in fostering harmony among divergent identities. In this context, a specific type of leadership and communication pertaining to identity assumes a crucial role.

In a study conducted by Moreno-Bella (2023), it was proposed that individuals' perceptions of economic inequality could potentially influence their perceptions of gender differences. The authors conducted two experiments in Spain ($N = 170$) and Mexico ($N = 215$) to examine the impact of high (vs. low) economic

inequality on the perceived agency and communion of men and women. The results of their study indicated that there was a decrease in communal content in social perceptions of both men and women when economic inequality was high compared to when it was low. In the context of elevated levels of inequality, there was an observed amplification in the perceived level of agency ascribed to men compared to women, while the disparity in perceived levels of communion between men and women diminished. In other words, men were perceived as even more agentic relative to their communal attributes, whereas women continued to be perceived as more communal than agentic, albeit to a lesser extent.

In order to determine whether categories were viewed as biological or social, Xu et al. (2022) investigated the impact of culture on the categorization process and the use of inductive reasoning. Using a triad picture problem, the authors of this study examined members of two very different cultures: those of the United States (which values individualism) and China (which values collectivism). There were a total of 581 people in the sample, distributed across four categories: Eight experiments were conducted on Chinese children, American undergraduates, and Chinese graduate students; two trials contrasted gender and occupation, two trials contrasted age and occupation, and four trials contrasted social class with age and gender. Using a combination of categorization trials, induction trials, and exploratory analysis, they discovered that the mechanism for categorising socially according to their stages of development differed from one culture to the next. Adults in the United States were found to use biologically relevant categories more frequently than children, while adults in China were found to use socially relevant categories more frequently than children and adults in the United States.

Ghosh and Huq (1985) conducted research on the social identity patterns of two distinct ethnic groups that belonged to India and Bangladesh. They evaluated the social identities of individuals who belonged to religious groups such as Hinduism and Islam using different frames of reference, including self, own group, and out-group evaluations that crossed national boundaries. They carried out two investigations that were connected, one with 150 male students from India and a second involving 180 male students from Bangladesh. The research included individuals from both religious groups. The researchers utilized an open-ended interview followed by an adjective checklist to evaluate all three frames of reference.

Then they used a discrepancy score between positive and negative evaluations to produce a 2×3 factorial design. After doing statistical analyses, they determined that self and own-group judgments were similar across the two countries and that Hindu and Muslim individuals exhibited similar patterns of social identification but that out-group judgments were not.

In the study conducted by Grigoryan (2020), a factorial survey experiment design was utilized, involving a sample of 524 individuals representing six distinct ethno-cultural groups in Russia. The majority of the participants in the sample were of Russian ethnicity. A comprehensive analysis was conducted by crossing a total of eight categorization dimensions. The present study investigated the patterns of evaluation that emerged in scenarios involving crossed categorization, wherein multiple real-world categorization dimensions were intersected. Furthermore, this research investigated the moderating influence of identity importance within cross-categorization contexts. The study's research design comprised of two distinct stages. Initially, expert interviews were conducted to ascertain the most relevant dimensions of social categorization within the cultural context under investigation. Subsequently, a factorial survey was undertaken, enabling the manipulation of social categories to examine the causal impact of the dimensions on attitude. Prior to evaluating ten vignettes, participants were requested to provide socio-demographic information. The participants were presented with a set of three questions, wherein one question assessed perceived similarity, while the remaining two questions gauged attitude towards the individual. Furthermore, the participants offered indicators pertaining to the perceived similarity, contact, and perceived conflict among different ethnic groups. The analysis of the factorial survey data employed a hierarchical data structure. The primary results of the study provided substantial support for the additive nature of crossed categorization. They expressed disagreement with the assumption that category-based information processing would not be employed in situations where there is a higher quantity of categories. The significance of identity was found to be a strong predictor of the level of intergroup bias, specifically in relation to the dimension of religion. It is worth noting that religion emerged as the most influential dimension within the context of this particular sample.

In order to study how prejudice and identity undermining can be a

consequence of in-group and out-group meta-prejudice, Putra (2014) recruited 214 respondents (127 females and 87 males), the majority of whom were Sunni Muslim residents of Jakarta and Bandung, ranging in age from 17 to 56. The minority groups studied were Ahmadiyya and Christians. In this study, a 5-item Muslim identity scale with a Cronbach's α of .75 and a threshold score of 3.5 was used to determine individuals' identification with their religious group. In addition, prejudice belief was measured using responses to a 6-item bipolar adjective scale that asked participants to express their opinions. Finally, a meta-prejudice scale with two measures that were used to indicate metrics of in-group and out-group meta-prejudice was also used. In order to measure identity undermining, participants ultimately filled out a modified version of Sindic and Reicher's (2009) Scale. The results of the study indicated that the influence of prejudice towards one's own group, known as in-group meta-prejudice, consistently mediated the relationship between prejudice towards other groups, known as out-group meta-prejudice, and the negative effects on prejudice and erosion of one's own identity. The results of the study indicated that the influence of prejudice towards out-groups on prejudice and identity erosion was consistently mediated by meta-prejudice towards in-groups. The results additionally indicated that how an in-group was perceived by an out-group may also influence how group members perceive a particular out-group in their group.

In another study conducted by Putra (2016) which looked at the circumstances within which in-group and out-group meta-prejudice acts as functions of prejudice. The purpose of this study was to test the hypothesis that self-evaluation within one's own group would serve as a moderating factor in the relationship between prejudice and in-group/out-group meta-prejudice. In this particular study, 37 individuals out of a total of 362 (199 Christians living in Maumere, Flores Island, and 196 Muslims living in Maumere) had to be excluded due to insufficient or missing data. Scales with high Cronbach's α reliability on samples were used for this research, including a 6-point prejudice belief measure, a meta-prejudice measure, and an in-group self-evaluation measure. The results of the study provided support in favour of the prediction of the research, which stated that in-group self-evaluation moderates the link between the impacts of in-group and out-group meta-prejudice on prejudice. Higher levels of both in-group and out-group meta-prejudice, as well as a more favourable assessment of one's group, were also identified as significant indicators of

prejudice.

In an effort to determine the degree to which perceivers believe people in their in-group and their out-group exhibit intergroup assessment bias and out-group homogeneity, Judd et al. (2005) carried out a number of experiments focusing on the structure and content of attributed intergroup beliefs. Researchers focused their attention on the intergroup context of ethnicity, gender, and nationality in their studies. Findings were provided based on the findings of three distinct datasets: the first dataset investigated attributed group belief, the second dataset investigated gender stereotypes, and the third dataset investigated nationality stereotypes. In all three experiments, a factorial mixed-design ANOVA was applied, and correlation coefficients were computed on a variety of different measures. According to a number of studies, they concluded that attributed beliefs were prone to start intergroup contacts on the wrong foot, and perceivers attribute to others more skewed intergroup beliefs than they actually hold.

Bettencourt et al. (1997) conducted a series of three experiments to examine the role of category-based expectancy violation in explaining the tendency to harshly evaluate individuals belonging to both in-group and out-group categories. The manipulation of expectancy violation was conducted in three distinct studies, wherein various descriptions of in-group and out-group targets were employed. The participants were instructed to evaluate the global favourability of either an in-group or an out-group target, as well as to assign ratings to the target's trait features. Additionally, they demonstrated the extent to which the targets deviated from the established norms established by the categories. The first experiment conducted at Midwestern University involved a sample of 115 undergraduate participants. This experiment consisted of two distinct situations that violated category-based expectations. The results of this study demonstrated that the violation of expectations had an impact on the overall evaluations of the target's favourability. Furthermore, it was found that the assessed expectancies played a mediating role in these evaluations. Experiment 2 was carried out at Midwestern University and involved a sample of 178 undergraduate participants, aimed to replicate the findings of Experiment 1. The results of Experiment 2 indicated that the evaluation of in-group and out-group objectives followed a similar pattern when category-based expectations were not violated. Experiment 3 involved the participation of undergraduate students hailing

from Midwestern University. The students were tasked with presenting job resumes that were either in violation of or adhered to stereotypical expectations associated with female and male job candidates. The experiment was conducted with 131 female students and 111 male students. The findings demonstrated that both female and male targets received a more favourable evaluation when they defied the stereotypical expectations associated with their gender group. Furthermore, these results were not affected in any way by the subjects' membership in any particular group.

Hässler et al. (2020) conducted a comprehensive study to investigate the correlation between intergroup contact and endorsement of social change initiatives targeting marginalized populations. A total of 12,997 individuals from four distinct demographic groups (ethnic majorities, cis-heterosexuals, ethnic minorities, and LGBTIQ+) residing in 69 different nations were included in the sample for this research investigation. Various methodologies were employed to assess the conceptual frameworks of intergroup contact and endorsement of social change. The researchers initiated the analysis by conducting a regression analysis, where the original items were regressed on the subsample identifier variable. This regression allowed them to obtain the residualized item scores. The researchers employed confirmatory factor analysis in order to determine the definitive set of items. Ultimately, the researchers employed specification-curve analysis as a means to assess the bivariate associations between intergroup contact and the endorsement of social change. Based on the findings of this study, it has been demonstrated by researchers that there exists a positive association between intergroup contact and support for social change aimed at achieving equal treatment among members of advantaged groups, such as ethnic majorities and cis-heterosexual individuals. Conversely, a negative association has been observed between intergroup contact and support for social change among members of disadvantaged groups, including ethnic minorities, sexual minorities, and gender minorities. Irrespective of the method employed to assess contact and support for social change, the examination of specification curves revealed considerable variability in the strength and occasionally the direction of the observed correlations. Through their analysis, it was ascertained that the manifestation of a collective inclination to collaborate in unity served as a type of assistance for facilitating transformative processes, exhibiting a positive correlation with intergroup interactions within both privileged and marginalized

societal groups.

Champoux-Larsson et al. (2022) undertook two experimental investigations with the aim of reproducing the findings reported by Baus et al. (2017). Specifically, they sought to examine the impact of language as a social category indicator on the recognition of faces within an old/new paradigm. In the first experiment, a total of 65 participants were recruited through advertisements and personal contacts within a university campus. However, 12 participants were excluded from the analysis due to methodological limitations related to their language proficiency. The researchers utilized a set of 80 pictures sourced from the Chicago Face Database and employed a set of 40 sentences that had previously been used in a study conducted by Baus et al. (2017). Additionally, utterances were presented in both Swedish and English to native Swedish speakers. The experiment comprised of two distinct phases, namely an encoding phase and a recognition phase. The presentation of stimuli was semi-randomized, with the use of counterbalancing, resulting in four different versions. According to signal detection theory methods for accuracy, there was no significant difference in the recognition of faces when they were presented alongside Swedish utterances compared to when they were presented alongside English utterances. In the second experiment, a different language pair was employed, involving a total of 27 participants who were recruited through various means, including advertisements and other methods of disseminating information. The researchers replicated the materials, design, and procedure utilized in study 1, with the sole modification being the substitution of English sentences with sentences in Spanish. Subsequently, the researchers proceeded to expose native Swedish speakers to a combination of facial expressions and verbal utterances in both the Swedish and Spanish languages. The recognition of faces was found to be more accurate when they were presented alongside Swedish utterances compared to when they were presented alongside Spanish utterances. Based on the empirical evidence from experimental studies, researchers have determined that language serves as an indicator of social categories. Furthermore, akin to other indicators of social categories, language can be influenced by multiple factors and is not inherently unconditional.

Tarrant et al. (2009) conducted a series of three experiments to examine the impact of social categorization processes on empathy towards individuals belonging to out-groups. A total of 370 students from Keele University were involved in the study. Experiment 1 involved a sample of 80 university students, consisting of 46 females and 34 males. These participants were asked to provide self-reported measures of their empathy towards a student who shared a distressing experience, as well as their intentions to offer assistance in response to this situation. The data collection process involved obtaining a transcript of a radio interview, as well as a set of 15 emotion labels. Participants were then asked to rate their emotional response on a 9-point scale. Following this, participants completed a questionnaire consisting of nine items that described different behavioural tendencies. A factor analysis was conducted on a set of 15 emotion items, followed by a mediation analysis. The findings indicated that participants exhibited higher levels of empathy and stronger intentions to help when the student belonged to an in-group university, as opposed to an out-group university. Experiment 2 comprised a total of 180 participants, with 135 identifying as females and 45 as males. Additionally, Experiment 3 involved 110 subjects, with 76 identifying as females and 34 as males. The findings of these experiments indicated that individuals exhibited heightened empathy towards out-group members when an in-group norm was activated, which specifically encouraged the experience of this particular emotion. The activation of this norm also resulted in the manifestation of more favourable attitudes towards the out-group, as observed in Experiment 3. Furthermore, the influence of empathy was found to completely mediate this effect. The results of this study suggest that, similar to other emotional responses, empathy is subject to the influence of social categorization mechanisms.

In their study, Ellemers and Barreto (2006) examined the impact of positive and negative expectations derived from social categories on individuals' self-views and emotional responses. In order to achieve the objective of the study, a stratified sampling approach was employed. A total of 463 individuals were selected from the database of an internet community managed by a national advertising community. The purpose of this sampling method was to obtain a representative sample of the Dutch population, taking into account factors such as age, gender, and education category. Fifty percent of the participants were instructed to recall a personal experience in which they were subjected to (unwanted) categorization that led to favourable self-perceptions (positive categorization), while the remaining fifty percent

were directed to recall an experience in which categorization resulted in unfavourable self-perceptions. This manipulation aimed to create distinct conditions of positive and negative categorization. In addition, participants were requested to compose a concise written account of a specific situation that was displayed on the screen. This was intended to assist them in maintaining awareness of the situation while completing the measures. The measures utilized in this study included Agreement with categorization, Detection, Emotions, Coping response, and Self-views. Agreement with categorization was assessed using six items with a Cronbach's α coefficient of .89. Detection was measured using three items with a Cronbach's α coefficient of .62. Emotions were evaluated using sixteen items adapted from the PANAS scale, with a Cronbach's α coefficient of .86. Coping response was assessed using nine items with a Cronbach's α coefficient of .70. Lastly, Self-views were measured using eight items with a Cronbach's α coefficient of .62. The results of this study indicate that positive categorization evoked a greater degree of positive emotions and agreement compared to negative categorization. Furthermore, individuals exhibited a reduced ability to perceive the external categorizations upon which positive self-expectations were founded, and demonstrated a decreased inclination to voice objections. Furthermore, the findings from meditational analyses indicated that due to impaired detection, individuals who were exposed to positive categorization exhibited lower levels of self-confidence compared to those who were exposed to negative categorization.

Cecilia and Aroo-Conaco (1993) employed a socio-cognitive approach, both methodologically and theoretically, in order to comprehend the notion of social categorization, which is posited as a prerequisite for the formation of social identity. The research was undertaken within the Philippine context, encompassing a cohort of 106 students aged between 16 and 26 years. The study was conducted at the University of the Philippines in Diliman, located in Quezon City. The participants self-identified their ethnic backgrounds and also classified themselves based on additional categorizations. The study involved the utilization of three questionnaires administered at three-to-four week intervals to assess participants' perceptions of their Filipino identity, perceptions of their ethnic group, and their overall social category memberships. The findings from the analysis of descriptive statistics revealed that national category affiliation is considered significant, although it is not as prominent as one's familial, gender, or religious group affiliations. The study found that

individuals generally held more favourable perceptions of their own ethnic group compared to the national group. Furthermore, there was minimal similarity observed in the characteristics attributed to these two distinct social categories.

The study conducted by Bigler et al. (1997) examined a range of hypotheses derived from intergroup theory to explore the impact of a newly introduced social category on the development of intergroup attitudes. This study involved the inclusion of 61 elementary school children, consisting of 27 girls and 34 boys, with ages ranging from 6 to 9 years. These children were administered the Multiple Classification Skill Test, as well as the Self-esteem scale. They were then assigned to one of three types of school classrooms, each characterized by a different approach to group assignment. The first type of classroom involved the functional use of "blue" and "yellow" groups, which were assigned based on a biological attribute. The second type of classroom also utilized "blue" and "yellow" groups, but the assignment was based on a random drawing. The third type of classroom did not have explicit groups, despite the presence of "blue" and "yellow" groups. Pre- and post-measures were administered over a span of four weeks, during which children were tasked with completing measures assessing their intergroup attitudes. The study's results indicated that the utilization of colour groups in a functional manner had an impact on children's attitudes towards members of these groups. The findings consistently demonstrated a bias among children, as they exhibited a preference for individuals within their own group. There is a positive correlation between the level of self-esteem in children and the extent to which they engage in intergroup stereotyping.

In a recent study, Lau (2021) conducted a comprehensive review that aimed to enhance our comprehension of individuals' political behaviour by reframing social categorization as latent structure learning. The review contends that in order to gain a more comprehensive understanding of how individuals construct representations of social categories and the subsequent influence on their behaviours, it is imperative to broaden the conceptualization of social categorization beyond mere dyadic similarity. In addition, it is imperative for a comprehensive understanding of social categorization to encompass domain-general, quantitative predictions that can be utilized to test hypotheses pertaining to this phenomenon. The researcher presents an alternative model that involves inferring latent groups of individuals through latent structure learning. Additionally, the researcher analyzed the experimental evidence

supporting this model and discusses its potential implications for comprehending the political mindset.

In their study, Connelly et al. (2016) conducted a comparative analysis of websites belonging to nonviolent and violent ideological groups, as well as non-ideological websites. The researchers aimed to investigate the influence of social categorization and moral disengagement on website credibility, while also exploring the associations between psychological processes and website credibility. The study employed the content analysis method to evaluate a total of 105 websites, which were categorized into three groups: 32 violent, 36 non-violent and 37 non-ideological. Prior to conducting the coding process, the researchers provided training to the raters on various dimensions, including social categorization, out-grouping, moral disengagement, credibility of content features, and credibility of structural features. The findings of this study revealed that websites associated with violent ideological groups exhibited a higher level of social categorization, out-grouping, and moral disengagement compared to websites associated with nonviolent ideological groups and non-ideological websites. The results of the regression analysis indicated that the three variables examined exhibited a negative relationship with the credibility of content and structure on websites associated with nonviolent ideological groups. However, these variables did not demonstrate a significant predictive effect on website credibility for violent groups or non-ideological groups.

The Contact Logger, a mobile application designed specifically for location-aware functionality, was introduced, and its usability was evaluated by Keil et al. (2020). This software facilitated the documentation of interpersonal and intergroup relationships taking place in various public and private settings, encompassing both indoor and outdoor environments and capturing the immediate circumstances in which they occurred. The primary benefit of this approach, in contrast to conventional methods, resides in its capacity to gather frequent and timely (near-time) self-evaluations of individuals' behaviours and experiences. Additionally, it facilitated the logging of geographical location data. The study involved the implementation of usability testing in an authentic setting, spanning duration of seven consecutive days. Throughout this period, a total of twelve participants were engaged, and they diligently recorded each instance of interaction they had with individuals belonging to an out-group, specifically older individuals. The application assessed an individual's

attitude towards an out-group by considering various factors such as the type of contact, gender and age of the individuals involved, duration of the contact, group typicality, relative or equal status, contact quality, and additional functions such as reminders, geographical location, and optional photos. Following the completion of the task, the participants were instructed to fill out a post-test questionnaire using pen and paper. This questionnaire aimed to gather information regarding the usability and overall user experience of utilizing the Contact Logger. The findings from the descriptive and qualitative analysis indicate that the application demonstrated viability and user-friendliness when compared to conventional measures.

Mistry and Shah (2020) examined religious prejudice and quality of life among Hindu and Muslim communities living in Ahmedabad city of Gujarat, India. Stratified random sampling was used to draw total 120 participants (60 Hindu, 60 Muslim); age ranged from 20 to 65 years and had same number of male to female participants. Participants' levels of prejudice and quality of life were measured by using standardized self-assessment scales. Results of two-way ANOVA suggested that Hindus had a higher level of prejudice and quality of life than participants of Muslim community and females scored higher than males on both the variables investigated in the study. Results also suggested that overall there was weak positive correlation between quality of life and religious prejudice against the other religion.

The study conducted by Wakefield et al. (2017) examined the relationship between life satisfaction and patient group identification using a cross-cultural methodology. The primary aims of the study were as follows: (a) to explore the correlation between patients' group identification and their level of life satisfaction, utilizing a diverse cross-cultural community sample; (b) to assess whether this relationship is influenced by the participants' nationality; and (c) to examine the potential enhancement of life satisfaction through the possession of multiple group identifications concurrently. The research was conducted using data from the first wave of a two-wave health in Group project. The study included a total of 1824 patients from Scotland and 2005 patients from Italy. Although the procedures differed between the two groups, the questionnaire used remained identical. A total of 3829 patients, consisting of 1642 males, 2184 females, and 3 individuals of unknown gender, with a mean age of 54.99 years ($SD = 16.17$, range 18–97 years), were included in the cross-cultural study. The researchers employed standardized scales to

assess group identifications, contact-intensive groups, and satisfaction with life. The results of this cross-cultural study indicated that individuals who exhibited a stronger sense of identification with each respective group were more likely to report higher levels of life satisfaction. The influence of nationality served as a marginal moderator in the association between family identification and Satisfaction with life. Specifically, the relationship exhibited greater strength among Italian participants compared to their Scottish counterparts. Additionally, a significant correlation was observed between group identification and life satisfaction, indicating that individuals who identified with a greater number of groups reported higher levels of satisfaction with their lives. These effects were observed while accounting for variables such as gender, age, employment status, nationality, and level of interaction with each group.

Livingstone et al. (2020) conducted a comprehensive investigation consisting of five distinct sets of experiments in order to explore the potential impact of the distinct role of felt understanding on intergroup relationships. The study conducted by researchers examined various intergroup outcomes, including action intentions, trust, support for political separatism (in Studies 1, 2, and 4), intergroup forgiveness (in Study 3), and evaluations of both the out-group and the relationship between the in-group and out-group (in Study 5). Various measures were employed, and surveys were assembled utilizing the EFS survey software developed by Unipark. The results indicated that in Study 1, which focused on Scotland-UK relations and involved a sample size of 5033 participants, and Study 2, which examined UK-EU relations and included 861 participants, the variable of perceived understanding consistently and significantly influenced various outcomes. These outcomes encompassed trust, intentions to take action, and political separatism. Notably, this influence extended to participants' actual voting behaviour in the 'Brexit' referendum, as observed in Study 2. The aforementioned effects persisted, even after accounting for out-group stereotypes and meta-stereotypes. In Study 3, it was found that the perception of understanding played a distinctive role in predicting trust and forgiveness towards out-groups. This was observed in the context of Catholic-Protestant relations in Northern Ireland, with a sample size of 1162 participants. Furthermore, in Study 4, which focused on Basque-Spanish relations with a sample size of 205 participants, the perception of understanding emerged as a significant predictor of political separatism. Notably, this relationship held true even after controlling for specific relational factors

such as negative interdependence and identity threat. In Study 5, a sample size of 190 participants was utilized to examine the impact of a direct manipulation of felt understanding. This manipulation was hypothesized to have significant effects on the evaluation of the out-group, as well as on perceptions of in-group-out-group relations. In general, the results presented offer consistent evidence supporting the significant impact of perceived comprehension on intergroup dynamics.

The study carried out by Grütter et al. (2021) investigated the adolescents' perceptions of parental expectations regarding friendships among peers from diverse socioeconomic backgrounds in Nepal. A total of 389 early adolescents, ranging from grade seven to ten, were included in the study. Among the participants, 53% identified as female. The sample demographic revealed that 53% of the participants identified as Hindu, while 43% belonged to various Nepalese ethnic backgrounds with distinct primary languages and customs. Additionally, 4% of the participants identified as Dalit, belonging to the lowest status castes. The measurement of participants' socioeconomic status (SES) involved the utilization of items pertaining to housing and property. Additionally, research assistants administered the Ambiguous Situation Task (AST) to participants, with an average duration of 20 minutes. The data underwent coding and subsequent analysis. The findings indicated that Nepalese adolescents, on the whole, ascribed greater negative intentions to peers from high socioeconomic status (SES) backgrounds compared to those from low SES backgrounds. Additionally, they anticipated that parents of high-SES individuals would express disapproval towards friendships formed across social groups, citing social hierarchies and reputation as reasons. The prevailing expectation regarding parents of individuals from low socioeconomic status (SES) backgrounds was that they would endorse and encourage the formation of friendships with individuals from higher SES backgrounds. This expectation was often justified by moral considerations and the potential for upward social mobility. The study also indicated that adolescents possessed an understanding of the systemic factors that contribute to socioeconomic status biases. The results of this study suggest that adolescents from low socioeconomic status (SES) backgrounds may face negative consequences when they are excluded from peer activities. To address this issue, parents can employ ethnic-racial socialization (ERS) strategies that aim to shield their children from the negative effects of SES disparities, limited social mobility, and instances of discrimination.

Doise and Sinclair (1973) distinguished between a “conflict and convergence of interests” approach of Sherif and a “categorization” approach of Tajfel in the area of the experimental study of intergroup relations. Researchers proposed their own point of view with an experimental study which aimed at comparing the images of group members in the form of their in-group and out-group situations where categorisation can take place and in situations where it is less likely to occur. 56 *collegiens* and 49 *apprentis* (age ranged between 15-17 years) were residing in Geneva, Switzerland participated in the study. Measure included the 12 item scale obtained from 15 lengthy interviews with *collegiens* and *apprentis*, out of 12 items only 6 items were used for statistical analysis. On the basis of results of these items four experimental conditions were made: (a) No encounter condition, (b) Symbolic encounter condition, (c) Individual encounter condition and (d) collective encounter condition. A factorial 2×2 analysis of variance (ANOVA) was conducted, and based on the results of the experiment, the researchers made the claim that the categorization approach can be used to elucidate intergroup phenomena and variations in intergroup representation.

The study conducted by De Cremer and van Vugt (1998) aimed to investigate the impact of salient collective identity versus personal identity on individuals' contributions in a public goods dilemma. Additionally, the researchers sought to identify the underlying processes that may explain this observed behavioural effect. A total of 93 undergraduate students from Southampton University were included in the study. The participants were divided into six groups, and the assignment to the experimental condition was done randomly. The participants were individually escorted to an experimental cubicle where they were positioned in front of a Macintosh computer. Participants were informed that they would receive feedback regarding the actions of their fellow group members through the computer interface. Subsequently, the trust of participants towards other target members was assessed, alongside their self-efficacy. The present study examined the effects of manipulation check, evaluative measurements, and contribution behaviour on individuals' identification with the collective and their level of contribution. The results indicated that participants exhibited a stronger identification with the collective and demonstrated higher levels of contribution when their collective identity was made salient, as opposed to when their personal identity was made salient. Moreover, it was

observed that this particular impact on behaviour was influenced by individuals' perceptions of their own ability to perform tasks successfully, rather than their perceptions of the reliability or credibility of their peers within the group.

In their study, Hall and Crisp (2005) conducted two experiments to examine the concept of intergroup bias reduction through the consideration of multiple criteria for social categorization. A preliminary investigation, known as a pilot study, was undertaken prior to the main experiments. The purpose of this pilot study was to assess the usability of a categorization task. This task required the participation of 45 undergraduate students who were tasked with considering alternative methods of classifying individuals, beyond the initial and prominent division between different groups. Experiment 1 involved the participation of 38 undergraduate students majoring in psychology, comprising 31 female and 7 male participants, with an average age of 19. These participants were assigned randomly to a mixed design with two factors: task (control vs. multiple unrelated alternatives) and target (in-group vs. out-group). The design also included repeated measures on the second factor. The procedure employed in the current study was maintained consistent with that of the pilot study, and the evaluations of the participants were assessed as the dependent variable. Experiment 2 involved the participation of 58 psychology undergraduate students (56 female, 2 male) with an average age of 19. These participants were randomly assigned to a mixed design with a 3 (task: control vs. multiple unrelated alternatives vs. multiple related alternatives) $\times$ 2 (target: in-group vs. out-group) factorial structure, where the second factor was subject to repeated measures. In Experiment 2, the procedure was maintained consistent with both the pilot study and Experiment 1. The dependent measure involved assessing the creativity, originality, and artistic merit of in-group and out-group paper models using a rating scale that ranged from 0 (*indicating no presence of the measured attribute*) to 9 (*indicating a high presence of the measured attribute*). The aforementioned items were aggregated to create indices measuring the evaluation of the in-group ($\alpha = .76$) and out-group ($\alpha = .81$), respectively. Based on empirical investigations, researchers have discovered that the generation of alternative social classifications, which are not connected to an initial target dichotomy, leads to a reduction in intergroup bias. This reduction in bias was observed when comparing the experimental condition to a control condition. Furthermore, the researchers were able to replicate this effect in a subsequent

experiment (experiment 2). Notably, the findings from experiment 2 indicated that the reduction in bias was only observed when participants generated unrelated categorizations, as opposed to related categorizations. The findings of this study also proposed the perspective that augmenting categorical complexity could serve as an effective strategy for mitigating bias.

The study conducted by Rhodes et al. (2018) investigated the manner in which language confers significance upon otherwise arbitrary distinctions among individuals during the process of social categorization. Two experiments were conducted by researchers. In the first experiment, a total of 97 toddlers were selected to participate. The mean age of the toddlers was 32.05 months, with 44 of them being male and 53 being female. The selection process involved obtaining consent from the caregivers of the toddlers. The testing took place at the Children's Museum of Manhattan. The initial group of 64 children who took part in the study were assigned in a random manner to either the generic or specific condition. Following that, a further 33 children were allocated to a control condition without any labels. In order to facilitate comprehension, the researchers in experiment 1 aggregated the data from all three conditions and subsequently evaluated the hypothesis using a match-to-sample task. In the second experiment, the researchers conducted tests to explore the potential disruption of categorization caused by the specific and no label conditions. This was done in contrast to their previously proposed explanation, which suggested that the generic condition actually aided in the process of categorization. Additionally, the researchers aimed to investigate the specific features of the generic sentences presented in Study 1 that might have contributed to the facilitation of categorization. In the second experiment, a fresh cohort of toddlers ($N = 32$, comprising 15 males and 17 females) was incorporated, while maintaining an identical procedure to that of the first experiment. The experimental findings indicated that the utilization of generic language, which pertains to abstract categories, played a significant role in shaping the progression of social categorization. The acquisition of a new category by toddlers was observed when they were exposed to generic language referring to individuals who possessed a shared arbitrary perceptual feature. However, this acquisition did not occur when the toddlers were exposed to matched specific language, simple labels, or plural (but non-generic) language describing the same group of individuals. The aforementioned findings demonstrate the influence of nuanced linguistic cues on the

process of social categorization.

In their study, Presaghi and Rullo (2018) conducted three experiments to examine the primary hypothesis that social categories are mentally organized in a left-to-right manner on an in-group to out-group continuum when individuals are highly aware of their membership. Two experiments were conducted to test a sub hypothesis regarding the spatial representation of social categories. Experiment 1 involved 18 participants and was conducted online, while Experiment 2 involved 43 Italian University students and was conducted in a laboratory setting. The hypothesis being tested was whether the activation of social category membership would lead individuals to implicitly position in-group categories to the left of a mental line, while placing out-group categories on the far right of the same mental line. The observed spatial organization remained consistent even when stimuli were displayed on either side of the screen, and their explicit position was incongruent with the implicit mental spatial organization of social categories. This finding was obtained from Experiment 3, which involved a sample of 50 right-handed students and was conducted in a laboratory setting. In general, the findings of the study revealed differential processing of in-groups and out-groups. The findings were analyzed in relation to social categorization theory, spatial agency bias, which refers to the tendency in Western cultures to mentally associate the agent of an action with the left side and the recipient with the right side, and the SNARC (Spatial-Numerical Association of Response Codes) effect.

Jang and Walther (2019) conducted a web-based experiment to determine if social identification cues in user comments generate in-group bias. 184 participants viewed a simulated Yelp.com page with local business information and four users' comments. Based on study they stated that self-categorization theory, viewers adopted their in-group's opinions when they strongly connected with them. When the dominant message was about an out-group topic, viewers were more inclined to identify with their in-group. Finally, an out-group-typed key message aroused stereotyped views of the out-group's knowledge but did not predict viewers' sentiments, supporting the in-group identification theory.

Two experiments by Hogg et al. (1986) examined how in-group discrimination (positive uniqueness) boosts self-esteem. Experiment 1 involved 52 11-12-year-old Bristol schoolboys. The school conducted experiment 1 in two

sessions in which independent variables of salience (individual versus group) and discrimination (cooperation versus competition) were experimentally manipulated in a 2×2 factorial design with subjects ($N = 13$) randomly assigned to conditions. The dependent measures were Oakes & Turner's (1980) self-esteem measure and behavioural strategies in the allocation of points on decision matrices. Experiment 1 decoupled discriminating and salience and validated the positive distinctiveness theory. A second experiment was conducted with 90 Bristol University students in the salience $\times$ discrimination design with an additional independent variable of self-esteem (high versus low). Compliance could partially explain the data. Experiments 2 supported the positive distinctiveness hypothesis over the salience hypothesis, but compliance was still a possibility.

In a study conducted by Shkurko (2013), a meta-analysis was performed on a total of 33 functional magnetic resonance imaging (fMRI) studies. These studies were selected from reputable academic sources such as Google Scholar and PubMed. Selective activations were reported for different social groups in order to identify shared neural structures that are responsible for representing social structure relationships. The utilization of Activation Likelihood Estimation (ALE) analysis unveiled the involvement of specific brain regions, including the bilateral amygdala, cingulate gyrus, fusiform gyrus, right temporoparietal junction, and right insula, in the processing of diverse aspects related to social categorization. The activation of the amygdala has been observed to potentially influence the modulation of behavioural responses to stimuli that are subjectively deemed significant. The anterior cingulate cortex (ACC) may exhibit differential activation patterns depending on the nature of social cognition tasks. Specifically, a ventral region of the ACC has been linked to self-referential reasoning processes pertaining to individuals within one's own social group, whereas a dorsal region of the ACC has been implicated in the regulation of emotional responses towards individuals outside of one's social group. The right insula may play a role in modulating out-group avoidance behaviour. The Fusiform Gyrus (FG) seems to play a direct role in the process of social categorization by modulating social perception through top-down mechanisms. However, establishing a clear connection between the identified clusters and the relational dynamics of in-group and out-group proved to be challenging.

In their study, Stoller and Freeman (2017) examined the neural underpinnings of social categorization by employing a unique experimental approach that integrated functional magnetic resonance imaging (fMRI) with real-time hand tracking. The present study investigated the neural mechanisms underlying the process by which the brain transforms a wide range of social cues into distinct categorical representations. In this study, a group of 16 adult participants, consisting of both males and females, were involved in the task of categorizing faces that exhibited variations in both gender and racial typicality. During the process of categorizing atypical faces, it was observed that participants' hand movements were drawn towards the unselected category response. This indicates the extent to which these faces activated the opposite category in a parallel manner. The study encompassed three distinct 2×2 within-subjects design categorization tasks, namely gender (gender: female vs male $\times$ typicality: typical vs atypical), race (race: black vs white $\times$ typicality: typical vs atypical), and colour (colour: blue vs red $\times$ typicality: typical vs atypical). The utilization of Multi-Voxel Pattern Analyses (MVPAs) has yielded empirical support for the presence of neural patterns in the right fusiform cortex that reflect the co-activation of social categories. The level of attraction of the hand towards the response option of the opposite gender or race category was found to be directly related to the similarity of neural patterns with the average pattern associated with that category. This similarity, in turn, was associated with a higher level of activation in the dorsal anterior cingulate cortex. The findings of this study offer empirical support for the co-activation of opponent social categories within regions of the brain responsible for processing facial stimuli. These categories engage in competitive processes that may eventually lead to a stable categorization, facilitated by regions involved in monitoring and resolving conflicts. Therefore, the results presented a neural mechanism by which the brain can potentially interpret a wide range of social cues and organize them into meaningful classifications of individuals.

The experiments conducted by Lorenzoni et al. (2021) aimed to investigate the role of language as a dimension of social categorization through experimental methods. The primary objective of the study was to examine the potential utilization of language as a factor in social categorization, even in cases where multiple languages coexist within a shared sociolinguistic group. The memory confusion paradigm was employed by the researchers. During the initial phase of the task,

referred to as encoding, participants were exposed to a visual stimulus in the form of a face representing the speaker, while simultaneously being presented with an auditory sentence to listen to. Two distinct languages were employed, whereby half of the facial stimuli consistently corresponded to one language, while the remaining half was associated with the other language. During the recognition phase, participants were presented with all the faces on the screen and were required to determine which face corresponded to each sentence that was presented during the encoding phase. The experiments were conducted using the Ibexfarm platform, which is an online platform. The test was made accessible to participants through a hyperlink, and recruitment for Study 1 was conducted using the participant pool database of The Bilingual Mind research group. In contrast, for Study 2, a 'snowball' sampling method was employed, utilizing social media platforms. The study included bilingual individuals from two distinct bilingual communities, namely the Basque Country in Spain and Veneto in Italy. The communities in question employed two languages, namely Spanish and Basque in Study 1, and Italian and Venetian dialect in Study 2. Subsequent investigations were conducted to examine the potential influence of daily language exposure on the role of language as a cue for social categorization. The results of the study indicated that participants demonstrated a higher tendency to select a speaker from the same language group when they made an error in attributing a statement to the appropriate speaker.

The study conducted by Clement and Krueger (2002) aimed to examine the potential influence of social categorization on the phenomenon of social projection. In order to achieve this objective, three experiments were conducted involving a total of 301 undergraduate students. The experiments examined how social categorization influenced the extent of social projection within laboratory groups that were either minimally categorized or had value tags assigned to them. Additionally, the experiments investigated whether individual perceivers made judgments about either groups or only one. The categorization effect observed alterations in the group status of the perceiver, resulting in a tendency for most perceivers to project only towards their present in-groups, rather than towards past in-groups. The absence of out-group projection provided support for the anchoring hypothesis, which posits that self-referent information is only activated when it is deemed relevant to the specific judgment being made. The study did not provide support for the induction hypothesis,

which posited positive out-group projection, or the differentiation hypothesis, which posited negative out-group projection.

In their study, Gordijn et al. (2001) conducted an experiment to investigate the occurrence of anger among individuals in response to perceiving intentional and unfair behaviour from an out-group. This behaviour was specifically examined in situations where the negative consequences were directed towards others, but not towards the individuals themselves. The study posited that the manifestation of out-group behaviour elicits feelings of anger in the individual perceiving it, contingent upon the classification of the individuals affected as either belonging to their own group or to a different group. A sample size of 57 female undergraduates from the University of Amsterdam ($M_{\text{age}} = 21.26$, $SD = 3.39$) were included in the study. These participants were assigned to one of three conditions using a random assignment procedure. The experimental materials consisted of a narrative designed to elicit anger, as well as measures assessing the participants' levels of anger, anxiety, and happiness. These measures were rated on a 7-point Likert scale, with 1 indicating no presence of the emotion and 7 indicating a strong presence. Additionally, participants' perceptions of fairness were also assessed. The study involved inviting participants to the laboratory in groups of 10. Upon arrival, they were directed to individual computer stations. At these stations, participants were exposed to priming information that emphasized either the differences or similarities between themselves and the victims. Subsequently, participants were exposed to negative behaviour exhibited by individuals from an out-group. The findings of the study supported the hypothesis that when individuals observed unfair and intentional behaviour by an out-group that harmed others, they experienced greater *Anger* if they perceived the victims as belonging to their own group rather than to an external group. Furthermore, the perception of victims as either belonging to the in-group or out-group did not have an impact on anxiety levels. This implies that particular emotions, rather than solely negative affect, were influenced.

According to the hypothesis proposed by Oakes et al. (1991), individuals tend to perceive the social category membership of others as prominent when their attitudes align with the corresponding social categorization, thus using it as a means to describe and explain their behaviour. Fit was operationally defined as the extent to which individuals perceived a correlation between the attributes of others and their

group membership in a manner that aligned with established norms. Experiment 1 involved the participation of 80 subjects, aged between 16 and 17 years, comprising 49 males and 31 females. These subjects were selected from a school in Bristol and were assigned to experimental conditions in a random manner, with each condition consisting of 20 subjects. In the initial study, participants were exposed to tape-slide presentations featuring groups of six individuals. The composition of these groups, categorized as either "solo" or "collective," as well as the level of agreement within the groups, categorized as either "deviance" or "conflict," were manipulated in a 2×2 design. This design aimed to establish a correlation between gender and attitudes within the solo/deviance and collective/conflict conditions. In the second experiment, the participants consisted of 90 British undergraduate science students, comprising 51 men and 39 women. These individuals were selected randomly and assigned to different experimental conditions. They were then shown videos featuring a group of three arts students and three science students. The videos were manipulated to vary the normative consistency of an arts target person, labelled as either "consistent" or "inconsistent," as well as the pattern of agreement among the group, categorized as "consensus," "conflict," or "deviance." This manipulation was designed to create a normative correlation between faculty membership and attitudes in the consistent/conflict condition. The measurement of salience involved the assessment of stereotyping and attributions related to individuals' social category memberships. The results of both experiments indicate that social categorizations were activated in a manner that accurately and adaptively represented social relationships. Veridical perception is facilitated by their activation solely in instances where they align with the objective reality. Furthermore, the findings provided support for the proposed hypothesis.

In their study, Araya et al. (2002) conducted a series of three experiments to investigate the impact of inadvertently introduced constructs that imply self-control on the activation of stereotypes linked to immigrants. In the first experiment, a total of 46 students (comprising 18 men and 28 women, with an average age of 28 years) willingly participated. The study employed a between-subject design. The study utilized a 12-item scrambled-sentence task, which was administered to participants as a language ability assessment conducted by the Department of Linguistics at Uppsala University. The critical priming stimuli encompassed words associated with control

and self-control, while the neutral prime condition consisted of neutral words that were also employed to elicit immigrant stereotypes. At the commencement of the experimental session, participants were tasked with completing the Modern Racial Prejudice Scale (MRPS). In the second experiment, a total of 52 students (22 men and 30 women) with an average age of 25 years voluntarily participated. The design of experiment 2 was kept identical to experiment 1, including the stimuli and procedure. However, in this case, the participants did not complete the MRPS at the beginning of the experiment. Experiment 3 involved the voluntary participation of 69 students (31 men and 38 women) with an average age of 23.4 years. A factorial design with two factors, stereotype activation (activation vs. no activation) and prime words (control-related vs. neutral), was employed. The stimulus material used in this experiment was the same as in experiments 1 and 2, with the exception that participants in the stereotype-activation condition were asked to complete the MRPS at the start of the experimental session, unlike those in the no-stereotype-activation condition. Based on the results of three conducted experiments, it was observed that participants who were exposed to control-related words exhibited a higher inclination to rate the target behaviour in a positive manner compared to those who were exposed to neutral words.

Yogeeswaran and Dasgupta (2014) conducted a series of three experiments with the aim of determining the specific circumstances in which multiculturalism can have either positive or negative impacts on the attitudes and behavioural intentions of individuals belonging to the majority group towards ethnic minorities. The researchers made predictions and subsequently observed that when multiculturalism was conceptualized in abstract terms, emphasizing its overarching objectives, it led to a decrease in prejudice among White Americans towards ethnic minorities compared to a control group. Conversely, when multiculturalism was framed in concrete terms, highlighting specific methods to achieve its goals, it resulted in an increase in prejudice among White Americans compared to the same control group. These findings were obtained through Experiments 1 and 2. Based on the investigation conducted by scholars utilizing social identity threat research, it was discovered that when multiculturalism was conceptualized in abstract terms, the perception of diversity as a threat to national identity decreased. Conversely, when multiculturalism was construed in concrete terms, the perception of diversity as a threat to national

identity increased. This perception of threat subsequently contributed to the development of prejudice, as evidenced by Experiments 2 and 3. The influence of individuals' political orientation on the impact of multiculturalism construal's on prejudicial attitudes and intentions to socially distance was examined in Experiment 3. The effects were mediated by symbolic threats to national identity, rather than realistic threats to national resources. The experiments conducted collectively provide evidence regarding the impact of multiculturalism on intergroup outcomes, specifically examining the factors that determine whether the outcomes are positive or negative. Furthermore, these experiments shed light on the influence of political orientation in shaping prejudice and behavioural intentions towards ethnic minorities, elucidating the underlying mechanisms involved.

The study conducted by Jenkins et al. (2018) aimed to examine the predictive factors associated with individuals' behaviour towards members of diverse social groups. The researchers conducted a series of experiments to demonstrate the significant relationship between social perception and social valuation processes in elucidating individuals' behaviour towards members of diverse social groups. Initially, the researchers conducted laboratory economic games to examine the impact of varying stereotypes on resource allocation behaviour towards different groups. They discovered that the quantitative variation in stereotypes regarding the warmth and competence of these groups had a significant influence on the allocation of resources. The findings from computational modelling also demonstrated that these effects were mediated by the interplay between social perception and social valuation mechanisms. Specifically, the dimensions of warmth and competence had contrasting impacts on the preferences of the 304 participants for fair allocations of resources. The framework demonstrated its efficacy in accurately predicting behaviour towards individuals belonging to various social groups in multiple samples. Furthermore, it effectively extended its predictive capabilities to account for societal disparities observed in labor and education contexts, exhibiting a high level of precision and accuracy. The findings of this study underscored a shared set of mechanisms that connect social group information with social treatment, and demonstrated how pre-existing, widely held assumptions about various social groups can generate and perpetuate social inequalities.

Devine (1989) conducted an empirical investigation to examine the fundamental assumption derived from a theoretical framework centered on the differentiation of automatic and controlled cognitive processes implicated in the manifestation of prejudice. In Study 1, a sample of 40 introductory psychology students who identified as white was examined. The findings of this study provided support for the assumption made by the model that individuals with both high and low levels of prejudice possess equal levels of knowledge regarding cultural stereotypes. According to the model, the activation of stereotypes occurs automatically when an individual encounters a member of a stereotyped group or a symbolic representation thereof. Additionally, the model posits that suppressing prejudiced responses necessitates the exertion of controlled inhibition over the automatically activated stereotype. Study 2 consisted of a sample size of 323 participants who completed a seven-item Modern Racism Scale that was integrated within a series of items related to politics, gender, and race. In Study 2, the researchers investigated the impact of automatic stereotype activation on the assessment of ambiguous stereotype-relevant behaviours carried out by an individual whose race was not specified. The findings indicated that when participants were unable to consciously monitor the activation of stereotypes, individuals with both high and low levels of prejudice tended to provide evaluations that aligned with the stereotypes. Study 3 comprised a sample of 67 White undergraduate students enrolled in an introductory psychology course. The primary objective of this study was to investigate the cognitive processes of individuals with varying levels of prejudice through the utilization of a consciously directed thought-listing task. The findings indicated that individuals with low levels of prejudice demonstrated a tendency to suppress and restrain the automatic activation of negative stereotypes, even when in an anonymous setting. Instead, they consciously substituted these stereotypes with thoughts that reflected their non-prejudiced beliefs. Subjects with low prejudice exhibited a limited number of pejorative thoughts. The individuals' thoughts were inclined towards considering the significance of equality or challenging the cultural stereotype.

Bodenhausen and Peery (2009) conducted an extensive literature review which focused on the occurrence of social perception in various contexts characterized by volatility, uncertainty, complexity, and ambiguity. These conditions encompass dynamic contexts, situations with missing information, scenarios

involving multiple bases for categorization, and instances where the meaning of available cues is unclear. Consequently, it is imperative for research to examine how these factors may influence or modify fundamental mechanisms of social categorization.

The research conducted by Yang et al. (2021) introduced a quantitative dynamical-system model to examine the combined impact of cognitive and social processes in the context of social categorization. According to the model's predictions, both groups exhibit a tendency to establish a boundary that is more restrictive in nature compared to the midpoint of the spectrum. Consequently, every group perceives individuals located in the intermediate region of the attribute space as members of an out-group. The hypothesis was examined by researchers through the analysis of U.S. political survey data, which focused on the perception of political independents by registered party members. Additionally, previous experiments investigating the perception of faces with racial ambiguity were also considered. The findings of this study provided evidence in favour of the hypothesis.

In their seminal work, Perdue et al. (1990) undertook a series of three studies in order to empirically examine the proposition that the utilization of linguistic expressions denoting membership in either the in-group or the out-group (e.g., us or them) might inadvertently sustain intergroup biases at an unconscious level. In the first experiment, a total of 23 subjects participated. During the course of the experiment, nonsensical syllables were subtly linked with pronouns that denoted either an in-group or an out-group. The findings indicated that syllables associated with in-group identifying pronouns, such as "we," were perceived as more agreeable in comparison to syllables associated with out-group designations, such as "they." In the second experiment, a group of 30 undergraduate students participated. The study examined the impact of in-group and out-group designators, which were presented as masked primes, on the time it took for participants to evaluate trait adjectives that were subsequently presented. Notably, the participants were unaware of the group-designating primes during the evaluation process. In Experiment 3, a group of 32 undergraduate students participated. The results indicated that the masked prime had a facilitative effect on the participants' reaction times when presented with positive person descriptors. This effect was compared to the influence of the masked prime "they" and the masked control prime "xxx." The use of the pronoun "they" did not

have a significant impact on participants' decision-making regarding negative descriptions of individuals.

The study conducted by Van Bavel and Cunningham (2009) comprised of two experiments aimed at investigating the potential of self-categorization with a newly introduced mixed-race group to supersede automatic racial bias. Experiment 1 comprised a sample of 109 undergraduate students from the University of Toronto, while Experiment 2 consisted of 126 undergraduate students from the Ohio State University. The participants were assigned to a mixed-race group in a random manner, and it was observed that they exhibited more favourable automatic evaluations towards members of their own racial group (in-group) who were Black, as compared to members of another racial group (out-group) who were also Black. When comparing these assessments to individuals of Black and White racial backgrounds that were not associated with either group, it was found that this preference was motivated by a preference for one's own group rather than a negative perception of individuals from the other group. Furthermore, it was observed that both faces belonging to the out-group and faces that were unaffiliated with any group evoked an automatic racial bias, with a preference for White faces over Black faces. This finding indicates that automatic evaluations were influenced by both the immediate intergroup context, such as positive evaluations of unfamiliar in-group members, as well as the factor of race, resulting in a bias towards out-group and unaffiliated faces. The experimental results yielded evidence indicating that self-categorization has the ability to supersede automatic racial bias. Additionally, the findings suggest that automatic evaluations of individuals can vary both across different social contexts and within the same context.

The study conducted by Cañadas et al. (2015) aimed to examine the extent to which individuals rely on categorical or individual knowledge when making cooperative decisions within a modified version of the trust game paradigm. The study involved the participation of 26 undergraduate students from the University of Granada. These participants were required to make decisions regarding cooperation with unknown partners of both black and white ethnicities. The decisions were influenced by the anticipated reciprocity rates of the partners. The manipulation of reciprocity rates involved the association of three out of four individuals from a specific ethnic group (either blacks or whites) with either high or low reciprocity

rates. Meanwhile, the remaining member of the ethnic group was associated with the reciprocity rates of the other ethnic group, thus creating inconsistency. The results of the analysis of variance (ANOVA) indicated contrasting performance patterns between white and black partners. The participants exhibited a tendency to categorize their white partners, as evidenced by their consistent cooperative decision-making across all partners. This was observed in their equal levels of cooperation with both consistent and inconsistent white partners. However, the aforementioned effect was not observed in the case of black partners, indicating a proclivity to perceive them as individuals rather than as representatives of a group.

In their study, Marques et al. (1998) implemented a series of four minimal group experiments involving a total of 26 high school students in Experiment 1, and 247 undergraduate students in Experiments 2, 3, and 4, with 69 participants in Experiment 2, 46 participants in Experiment 3, and 132 participants in Experiment 4. The purpose of the experiments was to examine the hypothesis that assessments of groups and their individual members are influenced by separate evaluations made at both the within-group and intergroup levels. Based on the theoretical frameworks of self-categorization theory and social identity theory, scholars made a hypothesis that individuals belonging to a group are motivated to establish both intergroup distinctiveness and validation of their in-group norms. The study manipulated several factors across Experiments 1-3, including membership (in-group vs. out-group), status of group members (modal vs. deviant), and either accountability to the in-group or to the out-group, or the salience of group norms. The presence of accountability and the prominence of social norms led to an increase in the tendency to derogate individuals who deviated from the norms of the out-group, while simultaneously elevating the status of individuals who adhered to the norms of the in-group. Experiment 4 demonstrated that the process of within-group differentiation served to reinforce the phenomenon of in-group identification. The results of this study indicate that subjective group dynamics play a role in enhancing social identity when individuals evaluate typical and atypical members of their own group as well as members of other groups.

Gaertner et al. (2006) performed a series of four experiments to investigate the necessity of inter-aggregate comparison for the formation of groups and the development of positive in-group attitudes. This investigation was motivated by the

assumption derived from the meta-contrast principle, which posits that the presence of an out-group is required for the establishment of an in-group. The researchers facilitated the development of new social groups, both with and without a contrasting group for members to compare themselves to. They also manipulated factors within each group, such as the level of interaction or interdependence among members. Irrespective of the feasibility of conducting inter-aggregate comparisons, the presence of intra-aggregate factors contributed to an enhanced perception of entitativity within the aggregate and fostered positive attitudes towards the aggregate, such as social attraction and cooperation among its members. The present study employed a series of four experiments, each conducted with a diverse sample size of students enrolled at Texas A and M University. The study employed a between-subjects design to examine the effects of manipulating the presence of an out-group and intragroup procedural interdependence. The researchers introduced variations in the level of interdependence among in-group members while they engaged in the task of solving three word-search puzzles. The participants were situated in individual cubicles and were presented with a single puzzle at a time. They were given a time limit of 2 minutes to identify and encircle as many words as they could for each puzzle. Subsequently, the researchers proceeded to assess the perceived entitativity of the aggregate, as well as the positive regard exhibited towards the aggregate, specifically referring to the social attraction and cooperation observed among its members. The findings from the mediation analyses suggest that the intra-aggregate factors were conducive to the development of entitativity, which subsequently led to an increase in in-group regard. The data presented indicate that the formation of groups and the positive regard for one's own group originate within the group itself, without the need for comparison to a different social entity.

Bernstein et al. (2007) investigated whether identifying people as in-group or out-group members was enough to evoke a pattern of face recognition similar to the cross-race effect (CRE), even when perceptual familiarity with the stimuli was held constant. In the first study, 60 white Miami undergraduates were categorized as real-life in-groups and out-groups (based on university affiliation) using 2×2 mixed-model experimental design and repeated measure on the first factor. In Study two, 43 White Miami Undergraduates participated, but 4 were eliminated owing to methodological concerns. The stimuli were 80 grey-scale faces of White college-age

guys with neutral expressions. Experiment 2 was identical to experiment 1. Minimal target groups were experimentally created. Despite similar perceptual skill for in-group and out-group faces, in-group targets performed better in both studies. These results revealed that social-cognitive mechanisms of in-group and out-group categorization were enough to generate face recognition performance disparities.

In their investigation, Hugenberg and Bodenhausen (2004) conducted two studies to investigate the impact of perceivers' prejudice and targets' facial expressions on race categorization, specifically examining whether these factors bias categorization in stereotypic directions. The researchers put forth a hypothesis that racial prejudice would exhibit a stronger correlation with the inclination to classify racially ambiguous faces as African American, particularly in the context of hostile expressions rather than happy ones. The researchers garnered backing for this hypothesis by employing a speeded dichotomous categorization task. In Study 1, a sample of 20 European American undergraduates participated, and both implicit and explicit measures of prejudice were utilized as predictors for the categorization of happy and angry targets. The manipulation of the target's expression was conducted using a within-subjects design. The stimuli used in the study consisted of 15 male faces that were generated using the Poser 4t three-dimensional character animation software. In study 2, a total of 57 European American undergraduates participated, and the design of the study was replicated from study 1. Additionally, a rating-scale task was included in study 2. The findings of the study revealed that there was a significant association between implicit prejudice, but not explicit prejudice, and heightened sensitivity to the facial expressions of the targets. This relationship held true regardless of whether prejudice was assessed subsequent to the race categorizations (Study 1) or prior to them (Study 2). The study's findings indicate that individuals' implicit biases can have an impact on their categorization of racially ambiguous faces, particularly when these faces exhibit negative emotional expressions. The research findings indicate that individuals with higher levels of prejudice demonstrated a greater tendency to classify racially ambiguous faces as African American when those faces exhibited expressions of anger, in contrast to individuals with lower levels of prejudice. The research findings indicate that implicit bias has the potential to influence individuals' perceptions and categorizations of others, with a heightened impact observed when facial expressions align with racial

stereotypes. This study emphasized the significance of taking into account the influence of implicit bias in shaping the process of social perception and categorization.

The Pew Research Center (2021) conducted an extensive survey in India to examine religious tolerance and segregation. The study involved approximately 30,000 face-to-face interviews with adult participants, conducted in 17 different languages, spanning from late 2019 to early 2020. The findings indicate that individuals from various religious backgrounds in India express a strong sense of freedom to practice their respective faiths. Religious tolerance is regarded by Indians as a fundamental aspect of their national identity. Within the context of various prominent religious groups, a substantial majority of individuals express the belief that demonstrating reverence towards all religions is of utmost significance in order to embody the essence of being authentically Indian. Tolerance held significant religious and civic significance, as Indians collectively recognized the paramount importance of demonstrating respect towards other religious traditions as an integral aspect of their own religious affiliation.

The study conducted by Verkuyten (2007) investigated the correlation between group identification and intergroup relations, with a specific emphasis on the context of religion. The study investigated the religious and Dutch national group identification of Turkish-Dutch Muslims. The present study examined the association between identification and overall affective evaluations of various religious groups, including Muslims, Christians, Hindustanis, Jews, and non-believers. Additionally, the study explored the relationship between identification and the endorsement of Islamic group rights. The study comprised a sample of 217 individuals who identified as Turkish-Dutch. A total of 206 participants self-identified as Sunni Muslims on an open-ended question, and these individuals were included in the subsequent analyses. Among the individuals included in the study, 68.8% were identified as males, while 31.2% were identified as females. The study included individuals ranging in age from 16 to 62 years, with $M_{\text{age}} = 30.18$ years ($SD = 10.63$). The measures employed in this study encompassed the assessment of Muslim group identification, which was evaluated through the use of six items utilizing a 7-point scale. The scale ranged from 1, indicating a strong disagreement, to 7, representing a strong agreement. Similarly, Dutch identification was evaluated using four items on the same 7-point scale. To

assess the sentiments of religious groups, participants were administered a widely recognized instrument known as the "feeling thermometer." Additionally, eight items specifically pertinent to the Dutch context were employed to evaluate attitudes towards Islamic group rights in the Netherlands. The research findings indicated that Muslim identification tended to be more nominal, characterized by high or 'total' identification, rather than a continuous variable. Additionally, a significant number of participants exhibited low identification with the national group. The affective evaluations of religious out-groups exhibited predominantly negative sentiments, with a notable emphasis on unfavourable perceptions towards Jews and individuals who identify as non-believers. The correlation between Muslim identification and attitudes towards the religious in-group, as well as the support for Islamic group rights, was found to be significant and robust. The study found a positive correlation between national identification and attitudes towards religious out-groups. However, this relationship was only observed among individuals who strongly identified as Muslims, which provides support for the mutual intergroup differentiation model.

In their research, Tadmor et al. (2012) conducted a comprehensive investigation consisting of six studies. The primary objective of their research was to examine the positive impact of multicultural experience on intergroup bias. Additionally, they sought to explore the motivational mechanism responsible for these effects, specifically focusing on the role of epistemic unfreezing. The researchers discovered that exposure to multiculturalism resulted in a decrease in the endorsement of stereotypes in Studies 1, 4, and 6. Additionally, it was found to reduce symbolic racism in Study 5 and discriminatory hiring decisions in Study 2. The researchers additionally provided evidence that engaging in multicultural experiences through experiments resulted in a decrease in the individual's Need For Cognitive Closure (NFCC) in Studies 3 and 6. Furthermore, they found that the positive effects of multicultural experiences on reducing intergroup bias were completely explained by the lower levels of NFCC in Studies 4, 5, and 6. The positive impacts of multiculturalism were observed irrespective of the specific stereotype group under consideration (including African Americans, Ethiopians, homosexuals, and native Israelis), regardless of whether the multicultural experience was assessed or manipulated, and regardless of the demographic characteristics of the participants (including Caucasian Americans or native Israelis). These findings highlight the

resilience and durability of this phenomenon. The collective results of various studies have consistently shown that multicultural experience is a significant factor in promoting social tolerance, primarily due to its influence on motivated cognitive processes. In their comprehensive analysis, Cikara and Van Bavel (2014) conducted a thorough examination of recent studies investigating the psychological and biological determinants that may contribute to the establishment of social groups, as well as the dynamics of cooperation and conflict within human populations. The objective of this study was to combine the existing research on intergroup neuroscience with traditional theories of group processes and intergroup relations. The aim was to go beyond simply describing the impact of specific social out-groups on the brain and behaviour. Instead, researchers focused on the fundamental psychological processes that govern interactions between different groups. These processes include the formation and modification of our perceptions of "us" and "them" through social identification and the functional relationships between groups. The aforementioned approach underscored the fluidity of social identity and the reliance of intergroup relations on contextual factors. The authors posited that the proposed theoretical integration has the potential to facilitate the resolution of apparent inconsistencies in existing literature. Additionally, they suggested that it could offer a framework for comprehending the fundamental components of intergroup dynamics. Furthermore, the authors emphasized the emergence of numerous promising avenues for future research at the intersection of intergroup relations and neuroscience. Additionally, the review paper proposes that enhanced integration across various methods and levels of analysis will ultimately establish the basis for more effective interventions and social policy.

In a study conducted by Carr (2017), the researcher examined the impact of religious disclosures on interpersonal attraction. The study drew upon social identification theory to elucidate the attributions that arise from religious disclosures in the e-mail signature blocks of professionals. A total of 268 individuals enrolled in various courses at a mid-sized Midwestern university in the United States were selected as participants for this study. These participants were randomly assigned to one of three experimental conditions, namely exposure to a signature block containing a Christian, Islamic, or secular quotation. Following this exposure, the participants were asked to complete assessments related to social identification and perceptions of

professionalism. The findings of the study revealed that, in contrast to previous research, the act of simply revealing one's religious affiliation did not lead to an increase in attributions. Instead, attributions regarding the professionalism of the message sender were positively influenced by the receiver's social identification with the sender's religion.

In their study, Cairns et al. (2006) performed secondary analyses on data collected from two separate random samples of the population of Northern Ireland. The first sample consisted of 1046 participants in the year 2000, while the second sample involved 1000 participants in the year 2001. The primary objective of their research was to investigate the influence of in-group identity and religious group membership on the extent to which in-group and out-group affect are interconnected. The study incorporated various measures, such as age, gender, socio-economic status, religious affiliation, and in-group and out-group affect. To assess these affective measures, a single-item feeling thermometer was employed. Participants were requested to rate their own community and the other community (which was regionally defined as the religious in-group and out-group, respectively) using a thermometer scale ranging from zero to one hundred. The participants were instructed that lower numerical values corresponded to less favourable emotions, while higher numerical values corresponded to more favourable emotions. Both surveys yielded results that demonstrated a prevailing in-group bias, as evidenced by higher affect ratings on the feeling thermometer for the in-group compared to the out-group. The impact observed was influenced by the level of identification individuals had with their respective in-groups, as well as their religious affiliation (Catholic or Protestant). However, these moderating factors also exhibited variations based on the varying levels of sectarian tension between the years 2000 and 2001. In both years, individuals who strongly identified with their social group and adherents of the Protestant faith demonstrated a higher degree of favouritism towards their own group compared to individuals with weaker group identification and followers of the Catholic faith, respectively.

In their research, Johnson et al. (2012) undertook two studies with the aim of investigating the correlation between religiosity and intergroup bias. Study 1 involved a sample of 144 undergraduate psychology students who were administered concise assessments to gauge their levels of religiosity and spirituality, as well as their

attitudes towards in-groups that aligned with religious values (specifically Christians and heterosexuals), and out-groups that deviated from these values (namely atheists, Muslims, and gay men). Consistent with expectations, individuals who self-reported higher levels of religiosity and spirituality exhibited a positive correlation with holding more negative attitudes towards out-groups compared to in-groups. Study 2 involved the participation of 73 college students who were enrolled in introductory psychology classes. The primary objective of this study was to investigate the impact of priming methods on intergroup bias by activating cognitive representations of religion. The results of the study indicate that participants who were subconsciously exposed to religious words exhibited a notable increase in negative attitudes towards out-groups that violated their values, compared to their attitudes towards in-groups. This effect was observed to be significantly larger than the increase in negative attitudes exhibited by participants who were primed with neutral words. The shift in relative attitudes can be attributed to the concurrent rise in in-group favouritism and out-group derogation. The aforementioned effects persisted even after accounting for self-reported levels of religiosity and spirituality, as well as pre-existing attitudes towards the aforementioned groups, as determined through statistical analysis. Moreover, the absence of interaction effects between religious primes and self-reported levels of religiosity suggests that the influence of the religion prime was the driving force behind the observed effects.

In an effort to address the limitations of prior studies, Hunter et al. (1996) sought to examine the enduring association between self-esteem and intergroup discrimination. To achieve this, the researchers employed realistic groups, employed multidimensional measures of self-esteem, and assessed self-esteem both prior to and following the manifestation of intergroup evaluative bias. The study comprised a sample of 271 adolescents, aged 16, who were enrolled in secondary schools in Northern Ireland. The researchers found that there was a significant relationship between engaging in evaluative intergroup bias and the enhancement of self-image esteem among members of realistic groups, as indicated by the results of the factorial ANOVA analysis. The study revealed a noteworthy increase in six specific domains of self-esteem, as identified by the instrument employed. These domains include honesty, academic ability, verbal ability, physical appearance, religion, and parental relations. The exhibition of bias did not have an impact on global self-esteem. The

aforementioned findings underscored the significance of incorporating realistic groups and domain-specific self-esteem when endeavouring to evaluate the impact of self-esteem on intergroup discrimination.

In accordance with the guidance provided by the theoretical and empirical literature that has been previously examined and reviewed, four aims were identified to study in the present research *changing bases of social categorization and intergroup relationship*, those aims were to: (1) identify the dominant typologies of everyday discrimination and the role of social categorization, (2) examine how in-group and out-group identification predicting social distance towards members of out-group, (3) assess intergenerational effects on social categorization and intergroup relationship, and (4) understand the category salience and intergroup relationship in cross-categorization situations. The overview of each aim was provided separately along with their research questions (aim 1), objectives and hypotheses (aim 2, 3 and 4) that were formed based on theoretical and empirical evidences previously reviewed.

1.6 Overview of Aims

1.6.1 Aim 1

The human brain possesses an inherent inclination to categorize objects and concepts as a means of comprehending the complexities of the world. It has been found by researchers that individuals include an innate tendency to categorize one another based on various group memberships, including but not limited to race, academic status (i.e., student or instructor), social roles, and other social categories (Fiske et al., 1991; Stangor et al., 1992). According to Allport (1954), the act of classifying individuals into categories such as male (in contrast to female), elderly (in contrast to young), Black (in comparison to Asian or White), and similar distinctions is an example of social categorization. In the context of social interactions, individuals tend to categorize others into distinct groups in a largely unconscious manner, often without deliberate contemplation (Crisp & Hewstone, 2007).

The tendency to categorize individuals into distinct categories is frequently advantageous. Categorization is employed in specific contexts as a means to acquire information regarding the characteristics of individuals who are affiliated with

specific social groups (Lee et al., 1995). Another way in which the characterization of social categorization as a heuristic is accurate can be observed. At times, individuals are classified based on convenience rather than for the purpose of gaining deeper insights into their characteristics, as limited time or energy constraints may prevent a more comprehensive approach. Furthermore, the utilization of preconceived notions about individuals to formulate an impression of another individual may facilitate a sense of simplicity in our lives (Macrae et al., 1994).

The social categorization process has its advantages, but it also has drawbacks that we can't altogether avoid. One issue arises from the phenomenon of social categorization, which has a tendency to influence our perceptions in a manner that leads to an overestimation of the disparities among individuals belonging to different social groups. Simultaneously, this process also fosters a perception of greater similarity among members of certain groups, particularly those categorized as out-groups, than is truly representative. As a result of this tendency to overgeneralize, there is an increased probability that individuals will exhibit similar thoughts and behaviours towards all members within a given group. Stereotypes are "images in our heads" that we have of various social groups (Lippman, 1922). Social categorization affects our impressions of groups, such as the idea of out-group homogeneity. These ideas appear to be correct and natural, despite the reality that they are often distorted overgeneralizations that cause intergroup conflict (Hirschfeld, 1996; Yzerbyt et al., 1994).

Intergroup biases such as discrimination, prejudice, and stereotyping originate from the cognitive process of social categorization, which involves the division of individuals into distinct social groups. Consequently, these phenomena are not solely individual occurrences but also possess societal and structural dimensions (Link & Phelan, 2001; Pager, 2008). Discrimination, in the context of intergroup bias, can manifest either towards an individual within a group or towards the group as a collective entity. In contrast, stereotypes and prejudice are internal psychological processes. In other words, these characteristics are intrinsic to an individual. Stereotypes and prejudices can exhibit variations not only in their outward manifestation but also in the level of conscious awareness held by the individuals harbouring these cognitive processes. Throughout history, stereotypes and prejudice have been commonly understood as overt reactions. The aforementioned concepts and

perspectives are consciously acknowledged by individuals, and they can be deliberately influenced and strategically managed in their expression (Fazio et al., 1995). In contrast to the aforementioned explicit, conscious, and deliberate processes, implicit prejudices and stereotypes are characterized by a lack of awareness and the activation of ideas in an unconscious manner. The mere presence of the attitude object has the potential to trigger the associated stereotype and attitude automatically, even in cases where the perceiver lacks conscious awareness of this cognitive process. Despite the widespread utilization of implicit attitude and stereotype tests in contemporary research (Fazio & Olson, 2003), there remains an ongoing scholarly discourse regarding the psychological significance of these concepts. According to Karpinski and Hilton (2001), there is a contention that implicit measurements of bias primarily uncover deeply ingrained and "automatic" cultural associations rather than reflecting individuals' explicit attitudes. It has been argued that implicit and explicit measures assess a singular attitude, but they do so at distinct stages of expression, wherein social desirability concerns exert a more significant impact on overt expressions (Fazio et al., 1995). However, there is a belief among specific individuals that implicit and explicit measures represent distinct elements within a dual attitudes framework. Implicit responses are often seen as manifestations of "older" attitudes and stereotypes that have been superseded by more recent explicit forms of prejudice or inadequately replaced by individuals who promote egalitarian beliefs (Wilson et al., 2000). Alternatively, implicit measures may capture different components of attitudes, including emotional and cognitive aspects (Rudman, 2004).

Discrimination is a daily occurrence for many people and has a negative connotation in the context of intergroup relationships. In addition to implying the need to differentiate between social objects, this term also refers to the unfair and unjust treatment of persons on the basis of their ethnicity, gender, age, or sexual orientation, this is known as discrimination (APA, 2022). According to Allport (1954) the act of discrimination is defined as the refusal to grant "individuals or groups of people equality of treatment which they may wish" (p. 51). Discrimination was described by Jones (1972) as "those activities aimed to retain own-group features and favoured position at the expense of the reference category" (p. 4). Discrimination is usually motivated by irrational fears and a general lack of understanding also prejudice based on ideas of identity and the desire to have a sense of belonging to a

specific group underlie all forms of discrimination. It can take the form of overt hostility towards a group member or, more subtly, a less positive response than would be given to a member of the in-group under the same conditions. In present scenario the politics of scapegoating and intimidation are on the rise in many regions of the world. An ever-widening chasm is being created between different communities as a result of intolerance, bigotry, and discrimination. The politics of fear is driving people further apart as leaders continue to peddle poisonous language and blame particular sections of people for societal or economic issues. When other individuals have a distinct identity, this can rise to feelings of division and animosity, and it can even lead to a dehumanisation of those people.

The existence of discrimination against individuals belonging to marginalized groups, such as people of colour, women, sexual minorities, overweight individuals, and other low-status groups, has been extensively substantiated through various empirical studies. For instance, the Pew Research Center (2013) has provided compelling evidence of the income gender gap, while the US Sentencing Commission (2013) has documented the sentencing disparity between Black and White, men and women. India discrimination report 2022 by Oxfam India, (2022), it is impossible to link specific events to discrimination with absolute certainty so in this research we incorporated all events that could potentially elicit the subjective experience of discrimination and identified the dominant bases for unfair treatment in daily life. In their study, Major et al. (2002) proposed a conceptualization of attribution to discrimination, which consists of two fundamental components. The first component involves making a judgment that an individual's behaviour is influenced by their social identification or group membership. The second component involves making a judgment that the individual's behaviour is unfair or lacking justification. From a phenomenological standpoint, an individual who ascribes their treatment to discrimination holds the belief that they were subjected to unfair or unjust treatment due to their affiliation with a particular social category or group. According to Major et al. (2002), instances that do not involve an individual's social identity, regardless of whether they are warranted or unjust, are improbable to be ascribed to discrimination. Likewise, circumstances that are perceived as justified or equitable are improbable to be associated with instances of discrimination. Even if individuals perceive treatment

to be administered on a collective basis, they are unlikely to label it as discriminatory if they perceive it to be equitable or acceptable.

According to Major et al. (2003), incidents are more likely to be perceived as instances of discrimination when two conditions are met. Firstly, the incidents must be associated with social identity. Secondly, they must involve unequal, unjust, or unjustified factors that either enhance the salience of group membership or provide injustice as a plausible explanation for the behaviour. These conditions tend to amplify attributions to discrimination. The propensity to ascribe an occurrence to discrimination is affected not only by the event's attributes but also by the person's qualities. For instance, a woman could blame her failure to advance in her career, in addition to discrimination, on her lack of managerial abilities, her inability to control her diet, an altercation she had a few months ago with her boss, or on the other candidates' apparent charisma and charm. There are rarely any objective criteria available to determine if discrimination has occurred. Because of this, discrimination evaluations are typically subjective, prone to human mistakes, and controversial. Nevertheless, attributing discrimination (or failing to do so) can have serious implications for one's mental and physical well-being, career prospects, and interpersonal relationships. Attributions of discrimination are complicated judgments that are formed not only by qualities of the event itself, but also by characteristics of the context, the culture, and the individual. When people believe that they or others have been treated unfairly based on their membership in a particular group, they are more inclined to attribute any behaviour to discrimination. Consequently, considering aforementioned premises the *aim 1* can be reiterated as '*identifying the dominant typologies of everyday discrimination and the role of social categorization*'. The research questions of this aim were addressed through the collection of *qualitative data*.

1.6.2 Aim 2

When people categorize one other according to characteristics like gender, religion, or ethnicity, they create two distinct social groups: their own, to which they belong, and an out-group, to which they do not. Individuals have the capacity to acquire information pertaining to their own identities and their position within society by means of categorization. Consequently, this process facilitates the development of a more comprehensive comprehension of one's self and the differentiation from

others. Individuals belonging to particular social categories do not perceive themselves or others as separate entities. Instead, individuals perceive themselves and others based on their membership in specific categories, and they identify themselves as part of those particular groupings. The degree of significance and relevance of category membership directly influences its role as a fundamental basis for individuals' social identities, which is a crucial aspect of their self-concept and a contributing factor to their group affiliation.

According to the social identity theory, people's subjective identification with various social groupings influences how they perceive themselves. Tajfel (1972) described a person's social identity as his "awareness that he relates to particular social groupings, as well as some emotional and value importance to him of group membership" (p. 292). This identification was considered to be more personal because it was backed by the individuals' self-categorization (Turner et al., 1987; Turner et al., 1994). The concepts of "identification" and "identity" are sometimes employed synonymously, as if they were referring to the same kind of social process. In Sigmund Freud's *Group Psychology and Analysis of Ego*, published in 1914 (republished by Institute of Psychoanalysis in 1959) the word "identification" first emerged. According to Freud's definition, it is a mode of interaction between individuals and groups within society as well as an unconscious relationship between a child and their parent. He viewed identification with a group leader as the foundation of group formation and one of the hidden forms of the Oedipus complex. At this point in time, the vast majority of academics examine identification as a method of individual socialization that affects the development of consciousness in an individual.

Identification is an on-going, unsettled, and open-ended socialization process. It encourages one to explore proactively and independently for one's persona and intensifies the subjective component in constructing a self-conception. The determining factors in this identification process include culture and social reality. Identification, therefore, combines emotional and cognitive aspects. In most cases, the development of the emotional component occurs during the early stages of a favourable relationship or as a direct consequence of positive feelings. Emotional connections form the basis for the cognitive component, ultimately resulting in adopting specific values, ideas, attitudes, and worldviews. As soon as an individual

realizes that his or her perspective and judgment of the world are similar to those of other members of the group, that individual experiences a more powerful positive emotional bond with the group. Positive and robust social group identification is linked to numerous positive consequences, such as enhanced coping, self-esteem, social support, and clinical outcomes, including reduced risk of depression (Haslam et al., 2009; Hughes et al., 2015; Paloutzian & Park, 2013). Hence, identification serves two primary purposes: (i) to satisfy the desire to be part of a community that offers safety and support and (ii) to integrate an individual into an established network of interpersonal connections.

Social categorization theory states that there are three stages to the identification process. In the first, people decide whether or not they want to be part of a group; in the second, they acquire the group's rules and stereotypes; and in the third, their group categories shape how they interpret and respond to all situations. Depersonalization occurs when an individual's salient social identity is given too much weight. This process increases the sense of similarities with other individuals of the in-group and differences with members of the out-group while at the same time reducing the degree to which an individual differentiates themselves from other people. This theory demonstrates that the "collective self" is responsible for explaining the majority of group occurrences, and why individual interests do not play a prominent part in group dynamics.

The smallest and most primitive communities consist of tightly knit families or tribes. It was one's kinship that determined whether they were part of the in-group or the out-group, and the first thing that had to be done when two strangers met was to determine their relationship with one another. If kinship could be demonstrated between them, they would be friends, and both would be considered members of the in-group. If there was no way to establish a relationship between them, then in many communities they were considered enemies and responded as such. People in today's modern culture are a part of so many different groups that the boundaries between their "in-group" and "out-group" relationships often become blurred. For instance, there are numerous in-groups that exist within a hostel, and these in-groups consider other people to be members of out-groups. On the other hand, when the hostel competes in a football game against the other hostel, all of the residents would act as an in-group and support their team on the field. In-groups and out-groups, as well as

identifying these groups, are significant because they influence behaviour. We have come to anticipate acknowledgment, loyalty, and helpfulness from our fellow members of an in-group, whereas our anticipations of the out-group change depending on the nature of the out-group. We anticipate antagonism from some other-group members, friendly competition from others, and apathy from still others. We may not anticipate hostility or apathy from the same out-group, yet our behaviour is unquestionably different.

Our actions change depending on the nature of the in-group or out-group to which we belong. It's also clear that the pronouns "we" and "they" and the associated attitudes towards in-group and out-groups are arbitrary constructions made up by individuals—members of the in-group tend to generalize and stereotype individuals of the out-group. As a result, residents of Delhi could have preconceived notions about people from Bihar or Uttar Pradesh (UP). The critical point is that such preconceptions are frequently developed by considering what appears to members of the in-group to be the least respectable features observed in members of the out-group.

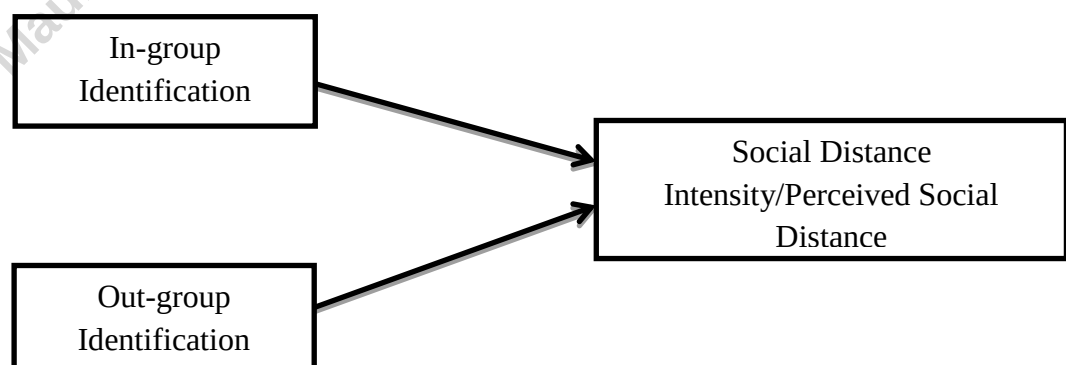
Self-categorization is also known as the act of defining oneself in relation to a social group, is the component of identification that is simultaneously the most fundamental and most important. For example, it is difficult to take pride in one's social group's accomplishments relative to those of another group if one does not see oneself as a member of that group. Evaluation is another significant aspect that plays a role in the identification process. This refers to the method by which a group is assessed compared to another relevant group on an essential feature. Depending on the results of the comparison, either a positive or negative evaluation can be given to the in-group. The importance of identification relates to whether group membership and the accompanying traits are fundamental to the self. One's level of attachment indicates how emotionally invested they are in a social group and how much one feels personally invested in the group's success or failure. When individuals have a strong sense of identification with the group, they are more likely to view things that are beneficial for the group as reflecting their personal preferences. Group members participate in various intergroup actions, such as collaborative action and social movement engagement, when group issues are regarded as simultaneously being personal preferences and goals (Simon & Klandermans, 2001). In reality, the idea of

social distance promotes such categorization and opposes individual uniqueness. The awareness of this principle helps to significantly lessen the negative consequences of categorizing into stereotypes and to break down the walls that prevent people from easily communicating with one another. In-group members are more likely to band together against out-group members in the face of any threat, whether actual or perceived. Robert E. Park (1924) was the first person to define the phrase and the notion of "social distance," which he described as "the grades and degrees of comprehension and intimacy which constitute personal and social relations generally" (p. 326). In contrast to geographical distance, social distance refers to the gap that exists between subsets of society. The concept encompasses all distinctions, such as social class, race/ethnicity, or sexual orientation, and the reality that the various groups do not interact. The Bogardus Social Distance Measure has been used extensively in social science studies since its creation in 1925. This measure is commonly used in the study of bias and intergroup relations because it allows respondents to open up about their level of tolerance towards people from different social groups. To understand the interplay of these variables researcher focused on *aim 2* which can be reiterated as '*to examine how in-group and out-group identification predicting social distance towards members of out-group*'.

Based on understanding developed over the review of literature concerned to variables under aim 2 the conceptual framework is shown as figure 1.1.

Figure 1.1

Conceptual Framework (Aim 2)



1.6.3 Aim 3

The issue of intergenerational relationships is an aspect of social life that has to be investigated using scientific methodology. Indian society is experiencing demographic shifts characterized by a rise in the proportion of younger individuals compared to other age cohorts, resulting in the emergence of a multi-generational modern Indian society. In contemporary society, there exists a coexistence of generations that exhibit a significant disparity in age, ranging from infancy to centenarians. These generations are characterized by distinct social contexts, shaped by diverse cultural and historical circumstances, and possess unique cognitive frameworks. The engagement between multiple generations is a significant aspect of contemporary individuals' lives, even in societies where their cohabitation is not customary (Easthope et al., 2017). These emerging patterns contribute to the escalation of communication challenges among individuals from varying generations.

The concept of generations can be examined through various viewpoints (Rudolph & Zacher, 2017). The idea of generations can be examined through multiple sociological, cultural, and psychological perspectives. Sociologically, generations are viewed as influential social forces that contribute to societal transformations (Mannheim, 1952). From a cultural standpoint, it is argued that generations actively shape culture rather than being solely shaped by it (Campbell et al., 2015). Additionally, the psychological perspective suggests that generations can provide insights into understanding human behaviour (Rudolph & Zacher, 2017). The concepts of replacement and temporal succession, which are fundamental to generational theory, have been applied in various contexts (Alwin & McCammon, 2007; Rudolph & Zacher, 2017). For instance, the concept of "generation as birth cohorts" refers to the idea that individuals are shaped by the historical period in which they were born. Another perspective is "generation as kinship," which explores how generational identities are influenced by familial relationships. Additionally, "generation as age-effect" examines how biological processes and lifespan development contribute to generational differences. Lastly, "generations as timing of entry into the organization and tenure" focuses on the succession of individuals within organizations (Joshi et al., 2011).

The concept of "generation as birth cohorts" holds significant importance within generational research, as it involves grouping individuals into cohorts based on

their birth years and attributing certain significance to these groupings (Rudolph & Zacher, 2017). According to Mannheim (1952), who is widely regarded as the originator of generation theory, individuals who belong to the same birth cohort share a collective position within the social and historical context. This shared position restricts them to a particular range of potential experiences, predisposing them to adopt a characteristic mode of thinking and engage in historically significant actions (p. 291). However, Mannheim's theory does not specifically focus on the effects of generation on individual behaviours and attitudes (Alwin & McCammon, 2007; Rudolph & Zacher, 2017). Scholars have highlighted that the shared experiences of individuals, such as parental influences, peer interactions, exposure to media, and engagement with popular culture, contribute to the development of collective memories during the formative years. Consequently, these shared experiences shape common value systems and behaviours (Lyons & Kuron, 2014; Parry, 2014). The enduring nature of cohesiveness in value systems and attitudes within a generation's lifespan has been demonstrated in studies conducted by Schuman and Scott (1989) as well as Arsenault (2004). Nevertheless, it is important to note that the impact of historical and social events on individuals during a specific time period may vary significantly (Alwin & McCammon, 2007). Lyons and Kuron (2014) and Parry (2014) have identified the presence of regional and cultural variations. The influence of value systems extends to various facets of an individual's life, encompassing their conduct within professional settings (Twenge & Campbell, 2008). Birth-year cohorts provide researchers with a means to analyze and describe potential variations in the trajectory of perspectives across different cohorts.

The social identity perspective aligns with the generational identity theory, as evidenced by the works of Lyons and Schweitzer (2017), Urick (2012), and Urick and Hollensbe (2014). The concept of generational identity refers to an individual's perception and understanding of their affiliation with a particular cohort of individuals within a given time period (Joshi et al., 2010). As previously elucidated individuals belonging to a particular generation exhibit a collective set of shared value systems. The presence of common value systems can influence the establishment of boundaries for social and self-categorization. This process involves individuals identifying with their respective generation, leading to the formation of a generational social identity. According to Tajfel's (1974) conceptualization of social identity, Joshi et al. (2010)

have defined generational identity as the awareness an individual possesses regarding their membership in a specific generational group or role, accompanied by emotional and value-based significance attributed to this membership. In conclusion, the generational identity theory elucidates the process of social categorization and intergroup dynamics that occur within and between generations, thereby enhancing our comprehension of generational phenomena. Hence, in order to gain a comprehensive understanding of the differences in in-group identification, out-group identification, social distance intensity and perceived social distance, aim 3 can be reiterated as follows '*to assess intergenerational effects on social categorization and intergroup relationship*'.

1.6.4 Aim 4

The phenomenon of social categorization, wherein individuals are classified as either in-group or out-group members, has been demonstrated to result in two enduring effects. Categorization, such as grouping individuals based on gender, has been found to result in the amplification of perceived similarities within groups (e.g., women perceiving other women as more similar) and differences between groups (e.g., perceiving increasing dissimilarity between women and men) (Doise, 1978; Doise et al., 1978; Tajfel & Wilkes, 1963). Furthermore, the occurrence of bias and discrimination can be attributed to the process of social categorization, wherein individuals are assigned to specific groups (Brewer, 1979; Doise, 1978; Messick & Mackie, 1989; Tajfel, 1978; Wilder, 1986). Indeed, research has demonstrated that the mere categorization of individuals into in-groups and out-groups based on insignificant differences leads to prejudiced assessments (Brewer & Silver, 1978; Doise et al., 1978; Kahn & Ryen, 1972) and the inequitable allocation of benefits (Tajfel et al., 1971; Turner, 1975).

The self-categorization theory under social identity theory has as one of its main tenets the responsiveness of social identity to immediate social situations. The cognitive system, driven by motives to reduce uncertainty and enhance self-perception, aligns social categories with characteristics of the social environment and highlights the category that is subjectively most significant in relation to the social context and one's personal position within it. The present study examines the relationship between category accessibility and category fit, focusing on individuals' tendency to rely on accessible categories and assess their compatibility with the social

context. The salience of the category that is most appropriate for the field is emphasized within that particular context (Oakes et al., 1994; Oakes & Turner, 1990). Categories can exhibit accessibility due to their perceived value, significance, and frequent utilization as components of one's self-concept (referred to as chronic accessibility), as well as their perceptual prominence (known as situational accessibility). Categories are applicable in the social domain because they consider the relevant similarities and differences among individuals in specific situations (referred to as *structural fit*). Additionally, categories are useful because they take into account the behaviours that are specific to a particular context (known as normative fit). Upon achieving complete activation, rather than being superficially tested, category specifications self-organize into prototypes that are contextually relevant. These prototypes serve as a foundation for emphasizing similarities within groups and differences between groups, ultimately enhancing distinctiveness and clarity. Self-categorization is a cognitive process whereby individuals identify themselves in relation to a specific in-group category that is currently salient. This process subsequently leads to a depersonalization of behaviour, as individuals align their actions and attitudes with the prototype or exemplar of the in-group. However, it should be noted that salience is not solely determined by the mechanical factors of accessibility and fit (Hogg, 1996b; Hogg & Mullin, 1999).

Based on Doise's (1978) category differentiation model, the process of single or simple categorization accentuates the distinctions between and similarities within categories, as proposed by Tajfel (1959). Conversely, the intersection of two categorizations results in "convergence" between the categories, thereby reducing the interclass effect, and "divergence" within each category, thereby diminishes the intra class effect. Therefore, as an illustration, the emphasis on the recognition of similarities within a specific category (e.g., Category A) will be mitigated by a concurrent emphasis on the recognition of differences, as Category A encompasses two distinct subgroups based on the X-Y categorization (Vanbeselaere, 1991). Numerous studies have demonstrated that intergroup discrimination rooted in the categorization of individuals as belonging to Group A or Group B can be mitigated or eradicated as a consequence of these processes. Deschamps and Doise (1978) demonstrated that the phenomenon of intergroup discrimination could be mitigated through the manipulation of categorizations. Specifically, they found that by

introducing a strong gender categorization and a weak, experimentally constructed red-blue categorization, the level of intergroup discrimination was reduced. Brown and Turner (1979) raised concerns regarding the methodology employed in this study. They criticized the lack of usage of categories with equal psychological significance, which could potentially result in the cancellation of effects. Additionally, they noted that subjects in different conditions were presented with tasks of varying cognitive complexity. For instance, primary-school subjects were required to simultaneously think in terms of two categorization dimensions, a task that might have been too complex and led them to disregard categorizations altogether.

Deschamps and Doise (1978) believed that the emergence of intergroup bias can be attributed to accentuation processes inherent in social categorization. In scenarios involving primary categorization conditions, where individuals can be sorted into in-groups or out-groups based on a single dimension (such as race, where individuals are either classified as in-group or out-group), the cognitive tendency to emphasize similarities within groups and differences between groups is likely to fuel intergroup bias. In the majority of circumstances, individuals can be categorized as either in-group or out-group members across various dimensions. It is not unexpected that when multiple dimensions come together to distinguish between in-groups and out-groups (for example, where all in-group members are Latino and Democrat, while all out-group members are white and Republican), the differentiation between categories becomes more pronounced and intergroup bias becomes intensified (Biernat & Vescio, 1993; Deschamps, 1977; Oakes et al., 1991). In alternative instances of crossed categorization, the dimensions of categorization intersect in manners that result in the creation of four subgroups through the conjunction of said dimensions. When considering the intersection of race (specifically Asian or white) and gender (women and men), we observe the existence of various categories, such as Asian women, Asian men, white women, and white men. In relation to a specific perceiver, such as an individual of Asian descent, individuals can be categorized as either in-group members based on both dimensions (referred to as double in-groups, for instance, other Asian men), out-group members based on both dimensions (referred to as double out-groups, for example, white women), or in-group members based on one dimension but out-group members based on the second dimension (referred to as mixed groups, such as Asian women and white men).

Deschamps and Doise (1978) proposed that in contexts involving crossed categorization, the processes of category accentuation that drive social categorization along each individual dimension come into conflict and negate each other. The phenomenon of accentuating similarities based on a single dimension, such as grouping men who are similar to each other, should be mitigated by the simultaneous inclination to emphasize differences along a second dimension, such as distinguishing between Asian men and white men. Likewise, the emphasis on distinguishing characteristics based on the initial dimension (such as the differences between women and men) ought to be mitigated by the inclination to highlight commonalities along the secondary dimension (for instance, the similarities between Asian women and Asian men). Furthermore, Brewer et al. (1987) proposed four alternative models to address situations characterized by unequal category salience. Therefore, in order to acquire an in-depth understanding of accentuation processes resulting from social categorization in various cross-categorization scenarios, aim 4 can be reiterated as '*to understand the category salience and intergroup relationship in cross-categorization situations*'.

1.7 Rationale of the Study

Extensive research has been conducted on the phenomenon of social categorization. The existing body of literature provides substantial evidence regarding the manner in which social categories serve to simplify the complexities of the social world. These categories tend to oversimplify the variations within a particular group, while simultaneously emphasizing the distinctions between different groups. Furthermore, it has been observed that social categories exhibit a notable resistance to evidence that contradicts or challenges their validity (Brown, 2000). Social categorization is a process that involves the division of individuals within a social context into distinct groups. These groups are typically classified as either the "in-group," to which the individual belongs, or the "out-group," to which the individual does not belong. This categorization is based on specific characteristics such as gender, religion, or ethnic background. The process of categorization allows individuals to acquire knowledge about their place within society, facilitating a deeper understanding of their own identity as well as distinguishing themselves from others. Individuals belonging to specific social categories tend to perceive themselves and others not as unique individuals, but rather in relation to their membership within

these categories. The significance and depth of category membership directly contribute to the formation of individuals' social identity, which is an integral component of their self-concept.

The process of social categorization serves as the fundamental basis for intergroup relations. The establishment of relations between groups is contingent upon the ability to categorize the social world into distinct groups (Dovidio & Gaertner, 2010; Hogg, 2001). The attitudes and behaviours of individuals within social groups are influenced by the nature of the relationships between these groups. When there is negative interdependence, it often results in intergroup conflict. Conversely, positive interdependence has the potential to foster harmonious relations between groups. The phenomenon of intergroup behaviour in minimal conditions can be elucidated through the theoretical frameworks of social categorization, identification, social comparison, and the motivation for positive distinctiveness.

Social categorization is subject to two historical problems. Initially, it is essential to acknowledge that the social categories employed to conceptualize various groups are undergoing transformation. Furthermore, it is essential to note that all human groups are subject to historical developments and are inherently dynamic in nature. The perpetual evolution of culture contributes to the perpetual obsolescence of conceptualizations of culture. The comprehension of culture has undergone significant transformations, and concurrently, cultures themselves have exhibited a lack of permanence. The issue of the meeting, blending, and governance of various groups has been a longstanding concern in human history (Vertovec, 2010). Consequently, cultural diversity has existed since the inception of humanity (Kymlicka, 2010). Categorizing individuals based on their social affiliations is a complex task due to the fluidity with which individuals' transition between different social categories. Individuals traverse various social categories as they navigate through different locations, societal positions, developmental phases, gender identities, physical capacities, socioeconomic strata, and even cultural contexts. Individuals demonstrate their commitment to a novel nation by pledging their loyalty, embracing unfamiliar cultural practices, assuming additional obligations, or encountering altered perceptions. According to Riesch (2010), even if individuals do not physically relocate cultural elements such as artifacts and boundary objects have the ability to move, resulting in interactions and the repositioning of individuals. The underlying

premise is not solely that individuals belong to various social categories, encompassing both subordinate and superordinate categories, but rather that individual's switch between these social categories. In the context of the social structure in which we reside, individuals maintain various personal, ethnic, religious, cultural, and other identities. Social identities are ingrained in our cognitive processes and play a significant role in shaping our self-concept and various domains of our existence. Over the past three decades, particularly in India, individuals have witnessed significant transformations resulting from globalization, westernization, advancements in education and technology, and the emergence of the internet. These factors have facilitated increased proximity among people, which could lead to changes in their social categorization patterns and intergroup relationships.

Based on a growing body of evidence pertaining to the evolving nature of society and the increased interaction among individuals from diverse cultural backgrounds and social categories, it can be deduced that the foundations of social categorization are not static, but rather flexible and susceptible to change. This ability to change may result in the emergence of new patterns of intergroup relations, including the relative positioning of various groups within the social structure. This positioning refers to whether a particular group holds a dominant or subordinate position within society. Additionally, the fulfillment of other critical individual and collective needs may also influence these intergroup dynamics (Correll & Park, 2005).

Initially in India the intergroup relations were more rigid with problems like untouchability, religious social distance and others, but now majority of people of different groups go to same school, college, restaurants, work place, parks, living in same apartments even in villages people are familiar with urban culture because there is substantial increase in contact between people and enhanced information about other groups, considering these things the current research is an attempt to revisit and understand the '*changing bases of social categorization and its associated intergroup relationship*' (group identifications, and social distance) after engaging individuals in social categorization process based on their social category preferences. The present study was investigated by a researcher using an exploratory sequential mixed-method research design. In first phase of present research researcher focused on exploring the new and existing bases of social category dimensions along which individuals

experienced the negative intergroup behaviours, then in second phase which was built on findings of first phase researcher focused on understanding the various components and domains of social categorization and its influence on intergroup relationship based on individuals preferences of social categories and also examined their intergenerational effects on social categorization. The second phase of the present research also aimed at understanding the salient importance of social categories in cross-categorization conditions.

1.8 Research Questions and Objectives

Research questions and objectives have been formulated to direct study design, data collection, and analysis based on the guidance provided by the theoretical and empirical literature evaluated and the conceptual framework built for the study.

Research Questions [Aim 1]

Main Questions

1. What are the most dominant typologies along which people face discrimination in everyday life, and how are they distributed across individuals' demographics? 2. Also, how do these typologies interact to provide the bases for unfair treatment that people receive in everyday life?

Sub-questions

- a. What are dominant typologies of discrimination in peoples' daily lives?
- b. How do these typologies vary according to individuals' gender (male/female), age (young/old) and area of belongingness (urban/rural)?
- c. How do the different typologies of discrimination come together to provide the grounds upon which people are unfairly treated?

Objectives [Aim 2, 3 and 4]

1. To identify how social categories are distributed in terms of their preferences.
2. To assess the association between in-group identification, out-group identification, and segments of social distance (perceived social distance and social distance intensity) in participants' preferred social category.
3. To see in-group identification and out-group identification as predictors of social distance intensity in participants' preferred social category.

4. To see in-group identification and out-group identification as predictors of perceived social distance in participants' preferred social category.
5. To see the mean differences on in-group identification, out-group identification, and social distance intensity across participants' preferred social category in whole sample.
6. To see the ranking of perceived level of social distance across participants' preferred social category in whole sample.
7. To see the intergenerational effects on in-group identification, out-group identification, and social distance intensity in participants' preferred social category across generational cohorts.
8. To see the intergenerational effects on perceived level of social distance in participants' preferred social category across generational cohorts.
9. Identifying the trends in ranking of social categories in generational cohorts based on perceived level of social distance.
10. To examine the mean differences on social identification among whole sample and participants' double in-group membership in religion-caste cross-categorization condition.
11. To examine the mean differences on social identification among whole sample and participants' double in-group membership in age-gender cross-categorization condition.
12. To examine the mean differences on social identification among whole sample and participants' double in-group membership in economy-education cross-categorization condition.
13. To examine the mean differences on social identification among whole sample in language-place cross-categorization condition.

Hypotheses [Aim 2, 3 and 4]

- H₁ : The distribution of social categories across the sample will be unequal in terms of their preferences.
- H_{2a} : There will be relationship between in-group identification, out-group identification, and social distance intensity in participant' preferred social category.

- H_{2b} : There will be relationship between in-group identification, out-group identification, and perceived social distance in participants' preferred social category.
- H₃ : Regression coefficient to predict social distance intensity based on in-group and out-group identification will be other than zero in participants' preferred social category.
- H₄ : Regression coefficient to predict perceived social distance based on in-group and out-group identification will be other than zero in participants' preferred social category.
- H₅ : There will be mean differences on in-group identification, out-group identification, and social distance intensity across participants' preferred social category in whole sample.
- H₆ : There will be differences in ranking of perceived level of social distance across participants' preferred social category in whole sample.
- H₇ : There will be intergenerational effects on in-group identification, out-group identification, and social distance intensity in participants' preferred social category across generational cohorts.
- H₈ : There will be intergenerational effects on perceived level of social distance in participants' preferred social category across generational cohorts.
- H₉ : The ranking of different social categories will vary on perceived social distance in generational cohorts.
- H₁₀ : There will be mean differences on social identification among whole sample and participants' double in-group membership in religion-caste cross-categorization condition.
- H₁₁ : There will be mean differences on social identification among whole sample and participants' double in-group membership in age-gender cross-categorization condition.
- H₁₂ : There will be mean differences on social identification among whole sample and participants' double in-group membership in economy-education cross-categorization condition.
- H₁₃ : There will be mean differences on social identification among whole sample in language-place cross-categorization condition.